Saving Kermit: Dynamic Assortment Planning in a Boiling Market

Eneko C. Clemente[1], Oleg A. Prokopyev[1], and Denis Sauré[2]

[1]Department of Business Administration, University of Zurich

[2]Industrial Engineering Department, University of Chile

Problem definition. A retailer planning its assortment under evolving customers' preferences faces a tension between retaining past sales information to guide its assortment selection (*remembering*) and discarding that information once it no longer reflects purchasing behavior (*forgetting*). Reminiscent of the boiling frog apologue, ignoring this tension may quietly erode the retailer's revenue as the offered assortment becomes misaligned with customers' preferences. ***Methodology and results.*** We treat evolving customers' preferences as a risk that the retailer plans for upfront. The retailer does so by first choosing a *risk profile*, namely a class of evolving customers' preferences against which robustness is sought. Given this profile, robust planning becomes a regret-minimization problem relative to a clairvoyant retailer that knows how preferences evolve. We show that restart-based strategies achieve robustness when the restart timing is tied to the retailer's risk profile. ***Managerial implications.*** To manage the risk of evolving customers' preferences, a retailer can first articulate a risk profile aligned with its risk appetite and tolerance for experimentation, and then use that profile to decide when past sales information should be discarded. Alternatively, a retailer can reduce that planning burden by using more adaptive approaches. Doing so, however, comes at the cost of greater operational complexity.

1 Introduction

The growth of e-commerce and data-intensive retailing has made assortment planning a central operational decision for modern retailers. Digital channels allow firms to adjust displayed products quickly and observe customers' choices at an unprecedented granularity, creating new opportunities to infer customers' preferences and tailor assortments accordingly (Caro et al. 2020). Yet, the same channels that facilitate experimentation also force retailers to be selective about the products they present. Homepages, search results, and other prominent pages can accommodate only a small subset of the products in the retailer's catalog. Hence, assortment decisions are inherently selective, making revenue critically dependent on keeping offered assortments aligned with customers' preferences.

Such alignment is inherently dynamic. Customers' evaluations of products, responses to information, and resulting choices may evolve over time. Indeed, purchasing behavior has long been viewed as a behavioral process rather than a fixed mapping from products to choices (Sheth 1967). As preferences evolve, so does the value of historical sales data. When preferences remain stable, older observations improve the precision with which preferences are estimated; when they change, the same observations may instead guide the retailer toward assortments suited to customers it no longer faces. However, much of the dynamic assortment planning literature assumes that preferences remain static and therefore overlooks the risk that historical sales cease to describe the purchasing behavior of the arriving customers. The *boiling frog* apologue illustrates this risk: gradual changes may go unnoticed until their consequences become severe. Likewise, a retailer that continues to rely on an outdated assortment plan may fail to recognize these changes and, in the end, be boiled by the market.

Retailers differ in their data systems and in how effectively planners use them. Some firms make assortment decisions using sales histories together with additional demand signals, while others still rely partly on managerial judgment (Caro et al. 2020; Rooderkerk et al. 2022; Vorotyntseva and Honhon 2026). The use of information systems, however, does not automatically translate into better assortment decisions. Indeed, decision support affects performance through the ability of planners to interpret and use such systems (Vorotyntseva and Honhon 2026). Consequently, we do not view evolving preferences as calling for a *universal* algorithmic prescription, as much of the non-stationary bandit literature does (Besbes et al. 2019; Wei and Luo 2021). We view dynamic assortment planning instead as a robust-planning problem shaped by the retailer's appetite for risk and experimentation capabilities.

Research question. Evolving preferences create a fundamental trade-off between *remembering* and *forgetting* for retailers who use dynamic assortment planning algorithms. Remembering means using past sales information to learn preferences and better guide the assortment decisions, whereas forgetting means discarding historical observations when they no longer reflect consumers' purchasing behavior. We study how retailers can manage this trade-off under two planning regimes.

Under the *first* regime, the retailer specifies a risk profile, namely a class of evolving preferences against which robustness is sought, and applies a simple strategy tailored to that profile. Under the *second* regime, the retailer avoids such upfront specification and relies on a more adaptive strategy. We compare the two regimes by the protection they provide and the experimentation burden they place on the retailer. The comparison identifies when upfront planning can turn policies designed for static preferences into robust strategies and when evolving preferences call for more adaptive experimentation.

Model. The retailer repeatedly selects assortments over a horizon T. In each period, a utility-maximizing customer arrives, observes the selected assortment, and either purchases one product or leaves without purchasing. The retailer does not know how preferences evolve during the horizon, so early sales data may become uninformative later on. We capture this uncertainty through a risk profile that specifies the evolving preferences against which the retailer seeks protection. The retailer then faces a sequential decision-making problem under uncertainty (Hannan 1957), in which estimating customers' preferences must be balanced with deciding when past sales information remains useful.

A *path* of preferences and the retailer's assortment decisions jointly determine the purchasing distribution in each period (Train 2009). Hence, we compare such paths through the sales-history distributions they generate using the Kullback-Leibler divergence (Cover and Thomas 2005). Because this comparison operates at the level of sales histories rather than model-specific parameters, it applies to both parametric and non-parametric random-utility choice models. This comparison induces a magnitude measure for each path, capturing how much preferences change over time through the purchasing behavior they induce. Distinguishing paths of preferences in this way allows us to define risk profiles covering two extremes: paths with a single change and arbitrary paths with bounded magnitude.

We evaluate the retailer's performance by comparing the expected revenue generated under a given *policy* (i.e., an assortment strategy) to that of a clairvoyant retailer endowed with perfect knowledge of customers' evolving preferences. The resulting difference, known as *regret*, is a standard performance metric in the online learning literature (Foster and Vohra 1999). Regret captures the *opportunity cost* associated with operating under incomplete information about preferences. To account for the risk induced by the unknown and evolving preferences, we formulate the retailer's objective as minimizing worst-case regret (with respect to the chosen risk profile) and study how this quantity scales with the number of customers T. This objective admits a game-theoretic interpretation: the retailer commits to an assortment strategy, after which the environment responds adversarially by selecting a path of customers' preferences, subject to the constraints imposed by the retailer's risk profile.

Under static preferences, dynamic assortment planning typically centers on the classical trade-off between *exploration* and *exploitation* (Lai and Robbins 1985), balancing the need to learn customers' purchasing behavior with the goal of maximizing immediate revenue (Caro and Gallien 2007). When customers' preferences evolve, exploration and exploitation no longer fully describe the retailer's learning problem because the value of each sales observation depends on how long it remains informative. The retailer must then decide not only which assortments to learn from, but also how long the resulting information should continue to guide future assortment decisions. We study how the retailer's risk profile and experimentation capabilities govern this remembering-forgetting trade-off.

Contributions. A growing literature now equips retailers with dynamic assortment planning policies that achieve regret guarantees under static preferences. These guarantees, however, do not necessarily protect retailers when preferences evolve. We show how to recover such protection without redesigning policies from scratch. The retailer starts from a policy $\mathcal{A}$ for static preferences and a risk profile, namely a class of evolving preferences chosen *a priori*. This profile then determines how $\mathcal{A}$ is turned into a robust strategy by specifying when the sales history used by $\mathcal{A}$ is discarded.

We evaluate the resulting strategies by their worst-case regret over the chosen risk profile. Our analysis separates this regret into two components. The first one is the regret that $\mathcal{A}$ incurs under static preferences. The second one is the additional price paid to make the strategy robust to the evolving preferences contained in the risk profile. Our contributions, summarized in Figure 1, are fivefold.

First, we develop a sales-history measure of information depreciation. Instead of measuring changes in a parameter space, as in Foussoul et al. (2023) for the multinomial logit (MNL) choice model, we compare paths of customers' preferences through the purchasing distributions they induce and, in particular, through the resulting Kullback-Leibler divergence between sales-history distributions (Cover and Thomas 2005). Our measure applies to arbitrary random-utility choice models,

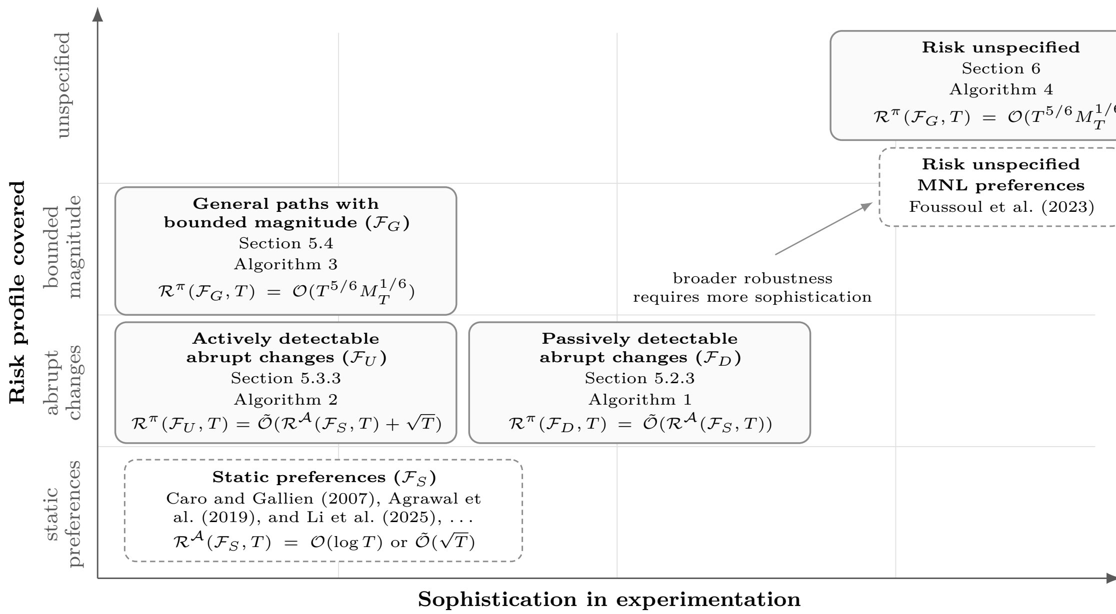


Figure 1: **Summary of our contributions.** Solid boxes indicate our robust-planning strategies and their regret guarantees under the specified risk profiles. Dashed boxes indicate the closest stream of literature. The figure highlights the trade-off between the level of robustness and the sophistication of experimentation required to attain it.

including parametric models such as MNL and non-parametric ones. We relate the measure to the MNL-specific variation measure of Foussoul et al. (2023): their measure controls ours, so their guarantees can be translated into guarantees under our measure, while the reverse implication may fail.

Second, we consider risk profiles that span two extremes in how customers' preferences may evolve: at one end, preferences undergo a single change; at the other, preferences may follow arbitrary paths, subject only to some control on how much they can evolve. We characterize these profiles along three dimensions. *Velocity* distinguishes abrupt changes from gradual movement, *magnitude* records cumulative changes in preferences, and *detectability* specifies whether changes appear in ordinary sales or only under targeted assortments. Together, these dimensions define a hierarchy of robustness, starting from *static preferences* $\mathcal{F}_S$. The profile $\mathcal{F}_D$ covers single changes that ordinary sales reveal *(passively detectable changes)*. The profile $\mathcal{F}_U$ covers single changes that may remain hidden unless some assortments are offered *(actively detectable changes)*. The profile $\mathcal{F}_G$ moves beyond a single change and allows paths of preferences whose magnitude is bounded by a prespecified *budget*.

Third, we connect these profiles to robust-planning strategies and characterize the corresponding attainable regret. The strategies differ in how they decide when to discard past sales information. For $\mathcal{F}_D$, passive monitoring lets $\mathcal{A}$ learn while using the assortments it selects to test whether purchasing distributions have changed, and restarts $\mathcal{A}$ after detection. For $\mathcal{F}_U$, active monitoring supplements $\mathcal{A}$ with a family $\mathcal{E}$ of assortments, tested periodically to reveal changes that the assortments selected by $\mathcal{A}$ may miss. For $\mathcal{F}_G$, the magnitude budget determines the restart interval; a fresh copy of $\mathcal{A}$ is run in each interval, and the accumulated sales history is discarded at every restart.

We quantify the lowest achievable regret under each profile and show that the proposed strategies match it up to logarithmic factors. For $\mathcal{F}_G$, periodic restarts attain this rate when $\mathcal{A}$ is robust to small changes, as we show for the policy of Foussoul et al. (2023). Without this robustness property, more conservative periodic restarts still provide regret guarantees, but they do not close the gap with the lower bound. Thus, achieving robustness depends not only on the retailer's risk profile, but also on the policy $\mathcal{A}$ for static preferences around which the robust-planning strategy is built.

Fourth, after characterizing strategies tailored to each risk profile, we propose an adaptive strategy for retailers that *do not specify* a risk profile in advance. The strategy serves as a comparison point for the targeted strategies above and illustrates the additional sophistication required when the risk profile is left unspecified. It requires no magnitude budget as input and is robust over $\mathcal{F}_G$. The construction follows the *bandit-over-bandit* approach of Cheung et al. (2022), building on corralling ideas for selecting among bandit algorithms without knowing in advance which algorithm best matches the environment (Agarwal et al. 2017). In our setting, the candidate algorithms are restarted copies of $\mathcal{A}$, each run with a different restart rate. The adaptive strategy learns how quickly to discard past sales by treating these restart rates as arms in a multi-armed bandit problem.

Finally, we evaluate the strategies numerically. Strategies tailored to a given risk profile perform well when realized preferences conform to that profile. The same strategies can perform well beyond their target profiles, even relative to more sophisticated alternatives, including our adaptive strategy and MASTER (Wei and Luo 2021) used with the policy of Foussoul et al. (2023). Their ability to do so depends on how the downstream policy responds to mild changes in preferences and whether the horizon is long enough for the benefits of adaptivity to outweigh its additional experimentation cost.

Managerial insights. Our results clarify the value of upfront risk planning in dynamic assortment planning with evolving customers' preferences. When the retailer can specify beforehand which evolving preferences it seeks to hedge against, robust-planning strategies provide simple responses that achieve robustness against the corresponding risk profile. Passive monitoring is appropriate for broad changes visible in routine sales. Active monitoring is needed when changes may remain hidden in routine sales and can be revealed only through targeted assortments. More generally, periodic restarts suit retailers that only commit to the view that older sales data may become less informative over time.

The same logic clarifies the limits of these strategies. Each strategy delivers a robustness guarantee only for the risk profile it is designed to cover. When the retailer cannot specify such a profile in advance, the risk profile must be built into the strategy rather than chosen upfront. The planning burden then shifts from selecting an appropriate response to designing a policy whose experimentation can discover which form of change is relevant. Robustness against evolving preferences therefore depends on the retailer's risk appetite and on the sophistication of experimentation it can realistically implement.

Organization of the paper. Section 2 reviews the literature. Section 3 formulates the dynamic assortment planning problem with evolving customers' preferences. Section 4 introduces a measure of variation in customers' preferences. Section 5 studies how risk profiles translate into robust planning.

Section 6 develops a more adaptive strategy for unspecified risk profiles. Section 7 reports the numerical experiments. Finally, Section 8 concludes. All proofs are relegated to the electronic companion.

2 Literature review

Customers' preferences. Customers in assortment planning are typically modeled as utility maximizers using discrete choice models (Mahajan and van Ryzin 2001; Talluri and van Ryzin 2004; Kök et al. 2015). These models take product-level utilities as primitives and map each assortment to a purchasing distribution (Train 2009). Much of the early literature adopts the multinomial logit (MNL) model for its analytical tractability (Rusmevichientong et al. 2010), though its independence of irrelevant alternatives limits the substitution patterns it can represent.

Later work addresses the restriction imposed by this property through richer choice models, including mixed logit (Feldman and Topaloglu 2015), nested logit (Gallego and Topaloglu 2014), and Markov chain choice (MCC) models (Blanchet et al. 2016). These models capture more realistic substitution patterns, but they also introduce estimation challenges. Mixed logit may overfit in data-scarce settings, whereas MCC models often lead to less tractable likelihoods, which complicates their estimation. Another line of work takes a non-parametric approach, representing customers' preferences as distributions over rankings of products (Honhon et al. 2012; van Ryzin and Vulcano 2015). We model customers as utility maximizers, but do not impose any particular discrete choice model.

Learning preferences. Because customers' preferences are unknown to the retailer, a large body of literature studies how to infer them from sales data. The literature typically distinguishes between offline and online learning. In *offline learning*, pre-existing sales data are used to calibrate a model of customers' preferences (Vulcano et al. 2012). For example, Farias et al. (2013) introduce a data-driven method that relaxes traditional parametric assumptions by inferring the model structure from sales data. In *online learning*, the retailer faces a sequential decision-making problem under uncertainty and uses its assortments both to generate revenue and to experiment, as each offered assortment produces a purchase outcome that, in turn, refines its preferences' estimates. We focus on online learning.

Multi-armed bandit (MAB) algorithms are often used in online learning to balance *exploration*, i.e., gathering information about customers' preferences, and *exploitation*, i.e., maximizing immediate revenue (Cesa-Bianchi and Lugosi 2006). Building on this idea, Caro and Gallien (2007) introduce an MAB approach to assortment planning. Subsequent work in dynamic assortment planning improves learning efficiency by exploiting the structure of the underlying choice model. Under MNL preferences, Rusmevichientong et al. (2010) obtain regret of order $\mathcal{O}((\log T)^2)$, later improved to $\mathcal{O}(\log T)$ by Sauré and Zeevi (2013), matching the rate of the asymptotic lower bound of Lai and Robbins (1985). These techniques have since been refined for several choice models, including MNL (Agrawal et al. 2019; Agrawal et al. 2026), nested logit (Chen et al. 2021), and MCC (Li et al. 2025).

Much of the literature assumes that customers' preferences remain static over the selling horizon. Because dynamic assortment planning policies designed for static preferences are few and highly specialized, we build on these advances by adapting existing policies (for most of our analysis) to evolving preferences through *restart* strategies. Restart timing is the central planning decision in our study.

Perishable information. The dynamic-pricing literature has long studied learning under initially unknown demand (Besbes and Zeevi 2009) and later extended the analysis to time-varying demand (Besbes and Zeevi 2011; Besbes and Sauré 2014; Keskin and Zeevi 2017). Dynamic assortment planning adds combinatorial and informational difficulties to this problem. Unlike a price, an assortment determines which products are shown to the customer. Sales reveal choices among offered products but do not directly reveal how customers would have chosen among excluded products.

Such censoring also limits the direct use of non-stationary bandit algorithms. Non-stationary MAB problems are typically handled by specialized algorithms based on restarts or sliding windows (Besbes et al. 2019; Wei and Luo 2021; Cheung et al. 2022). Applying these algorithms directly is computationally and statistically unattractive because the action space contains exponentially many assortments. Treating these assortments as independent arms would also discard the information shared across them through the choice model and generally perform poorly. While these ideas provide useful principles for designing specialized approaches, our goal is different: we seek to take existing policies designed for static preferences and make them robust to evolving ones.

A *black-box* reduction of this kind is not available; among existing approaches, Wei and Luo (2021) come closest to our goal. However, their reduction requires structural information from the base algorithm, in the form of an optimism certificate. In classical MAB, multiple algorithms satisfy such requirements. In dynamic assortment planning, such certificates are rarely available. *To our knowledge*, Foussoul et al. (2023) provide the only exception: they study evolving preferences under MNL and design a policy compatible with the reduction of Wei and Luo (2021). Existing reductions thus offer no *general* recipe for taking a policy designed for static preferences and making it robust to evolving ones. By contrast, the strategies discussed here are fully black-box precisely as a response to that limitation.

We view evolving preferences as a planning concern, not merely an algorithmic one. Retailers differ in their risk appetite and experimentation capabilities, which shape the strategies they can realistically adopt. Hence, the most adaptive strategy is not always the most appropriate planning response. Unlike in classical MAB problems, the question is not only how to adapt to a changing environment, but which form of protection the retailer can realistically implement against evolving preferences.

3 Problem formulation

Model primitives. We study dynamic assortment planning for a retailer offering $N \in \mathbb{N}$ products. Each product $i \in \mathcal{N} := [N] \equiv \{1, \ldots, N\}$ is sold at a price $r_i > 0$, yielding a unit profit $w_i \equiv$

$r_i - c_i > 0$, where c_i is its marginal acquisition cost. Customers arrive sequentially, one per period, and are indexed by $t \in \mathbb{N}$. Therefore, we use "time" and "customer" interchangeably throughout our study.

Each customer $t \in \mathbb{N}$ assigns a random utility U_i^t to each alternative $i \in \mathcal{N}_0 := \mathcal{N} \cup \{0\}$, where $i = 0$ denotes the *no-purchase* option. The joint distribution of these utilities, denoted by F^t, characterizes the preferences of customer t. We impose *no* additional structure on $F^{(\mathbb{N})} \equiv (F^t : t \in \mathbb{N})$, beyond requiring that the corresponding random utility vectors are defined on a common probability space, are independent across time, and satisfy $\mathbb{P}[U_i^t = U_j^t] = 0$ for all distinct $i, j \in \mathcal{N}_0$ and all $t \in \mathbb{N}$.

The selling horizon consists of T customer arrivals, indexed by $[T] \equiv \{1, \ldots, T\}$. Upon arrival, customer $t \in [T]$ is presented with an assortment $S^t \in \mathcal{S} := \{S \subseteq \mathcal{N} : |S| \leq K\}$, where K captures the retailer's limited display capacity. The customer then chooses $i^t \in \operatorname{argmax} \{U_i^t : i \in S^t \cup \{0\}\}$.

Assortment planning under full information. We assume that the retailer faces neither inventory constraints nor switching costs. These assumptions are commonly adopted in the dynamic assortment planning literature (Sauré and Zeevi 2013; Agrawal et al. 2019; Li et al. 2025) to isolate the cost of learning customers' preferences in terms of revenue. We define the purchasing probability for each product $i \in S^t \cup \{0\}$ from the offered assortment, including the no-purchase alternative, as $p_i(S^t, F^t) := F^t\big(\{x \in \mathbb{R}^{|\mathcal{N}_0|} : x_i \geq x_j \text{ for all } j \in S^t \cup \{0\}\}\big)$. For products that are not included in the assortment, we set the purchasing probability to zero, i.e., $p_i(S^t, F^t) = 0$ for all $i \in \mathcal{N} \setminus S^t$.

We let $r(S^t, F^t) := \sum_{i \in S^t} w_i \, p_i(S^t, F^t)$ denote the expected revenue associated with offering assortment S^t to customer t. For each customer t, we define the *revenue-maximizing assortment* as:

$$S^*(F^t) \in \operatorname{argmax} \{r(S, F^t) \, : \, S \in \mathcal{S}\}.$$

Throughout the paper, we assume that the revenue-maximizing assortment is uniquely defined.

Risk profile. In contrast to traditional dynamic assortment planning models, we allow customers' preferences to evolve over the selling horizon $[T]$. Thus, customers are described by a *path* of customers' preferences $F^{(\mathbb{N})} = (F^t \, : \, t \in \mathbb{N})$, rather than by fixed preferences satisfying $F^t = F^{t-1}$ for all $t \geq 2$. The retailer does not know the realized path of customers' preferences in advance.

Given this uncertainty, the retailer uses its available knowledge to specify a class $\mathcal{F}$ of evolving preferences that it may face. Such knowledge may take the form of a parametric choice model with parameters in a prescribed set, a substitution structure across products, or bounds on the frequency and size of changes in preferences. These restrictions define the retailer's *risk profile* and are made explicit later in Section 5. For now, we impose only a non-degeneracy condition. We assume that there exists $\varsigma \in (0, 1)$ such that every $F^{(\mathbb{N})} \in \mathcal{F}$ satisfies $p_i(S, F^t) \in [\varsigma, 1 - \varsigma]$ for all $i \in S \cup \{0\}$, $S \in \mathcal{S}$, and $t \in \mathbb{N}$. This condition rules out degenerate instances in which purchase outcomes become deterministic.

Planning under evolving preferences. Let $\mathcal{H}^t := \sigma\big((S^s, i^s) \, : \, s < t\big)$ denote the history of offered assortments and purchases prior to customer $t \in [T]$. A sequence of (possibly random) assortments $\pi := \big(S^{\pi,t} \, : \, t \leq T\big)$ is called an *admissible policy* or an *assortment strategy* if at each time $t \in [T]$, the assortment decision $S^{\pi,t}$ is a mapping that takes as input the past history $\mathcal{H}^t$ and returns a feasible assortment from $\mathcal{S}$. Specifically, $S^{\pi,t}$ represents the assortment offered by policy π to customer t. We denote by $\mathcal{P}$ the set of all such assortment strategies.

Following the literature (Cesa-Bianchi and Lugosi 2006), we measure the performance of any assortment strategy against that achieved by a clairvoyant retailer (called the *oracle*) with prior knowledge of $F^{(\mathbb{N})}$. Specifically, this oracle knows the preferences F^t of customer t, and therefore offers assortment $S^*(F^t)$ to that customer. For a given path $F^{(\mathbb{N})} \in \mathcal{F}$, we define the *oracle revenue* as:

$$J^*(F^{(\mathbb{N})}, T) := \sum_{t \leq T} r(S^*(F^t), F^t).$$

The oracle revenue defines the performance target against which the retailer's strategy is evaluated. It is the revenue the retailer would obtain with knowledge of customers' preferences, and is generally unattainable. Instead, an assortment strategy $\pi \in \mathcal{P}$ achieves the expected cumulative revenue:

$$J^\pi(F^{(\mathbb{N})}, T) := \mathbb{E}[\sum_{t \leq T} r(S^{\pi,t}, F^t)],$$

where the expectation is taken over the (random) assortments $(S^{\pi,t} : t \leq T)$ offered by strategy π.

Because the retailer does not know in advance which preferences path will be realized, we adopt a robust perspective with respect to a *risk profile* $\mathcal{F}$. The risk profile specifies the evolving preferences against which the retailer seeks protection by constraining the paths that the environment may select. Thus, weaker restrictions on $\mathcal{F}$ enlarge the environment's choice set, whereas stronger restrictions narrow it. We evaluate the retailer's performance relative to a clairvoyant retailer under the worst realization $F^{(\mathbb{N})} \in \mathcal{F}$. The *adversarial regret* of an assortment strategy $\pi \in \mathcal{P}$ with respect to $\mathcal{F}$ is defined by:

$$\mathcal{R}^\pi(\mathcal{F}, T) := \sup \{ J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) \ : \ F^{(\mathbb{N})} \in \mathcal{F} \}.$$

The adversarial regret quantifies the worst-case opportunity cost incurred by the retailer when committing to an assortment strategy under the risk profile $\mathcal{F}$. Given this profile, robust planning consists of constructing an assortment strategy that minimizes adversarial regret. Formally, we define:

$$\mathcal{R}^*(\mathcal{F}, T) := \inf \{ \mathcal{R}^\pi(\mathcal{F}, T) : \pi \in \mathcal{P} \},$$

as the lowest regret attainable by the retailer under the chosen risk profile. It captures the *price of robustness* associated with protecting against the evolving preferences in the risk profile $\mathcal{F}$.

4 Measuring sales-information depreciation

The retailer faces uncertainty about customers' preferences and how they evolve. The robust-planning problem introduced in Section 3 addresses such uncertainty through a risk profile, namely a class of evolving preferences against which robustness is sought. The risk profile determines how the retailer accounts for the depreciation of information contained in past sales data as preferences evolve. Constructing such profiles, however, requires a way to distinguish preferences paths from one another.

4.1 Comparing paths of preferences

In each period, the offered assortment and the preferences jointly determine the purchasing distribution; the resulting purchase outcome becomes part of the sales history that the retailer then uses to select its next assortment. Along a strategy $\pi \in \mathcal{P}$, two paths of preferences $F^{(\mathbb{N})} = (F^t : t \in \mathbb{N})$ and $G^{(\mathbb{N})} = (G^t : t \in \mathbb{N})$ can induce different distributions $\mathbb{P}^{\pi,T}_{F^{(\mathbb{N})}}$ and $\mathbb{P}^{\pi,T}_{G^{(\mathbb{N})}}$ over the entire sales history $\big((S^{\pi,t}, i^t) : t \in [T]\big)$. We compare these two paths through the Kullback-Leibler (KL) divergence $D_{\mathrm{KL}}(\mathbb{P}^{\pi,T}_{F^{(\mathbb{N})}} \,\|\, \mathbb{P}^{\pi,T}_{G^{(\mathbb{N})}})$ between the sales-history distributions they induce; see Cover and Thomas (2005).

The KL divergence $D_{\mathrm{KL}}(\mathbb{P}^{\pi,T}_{F^{(\mathbb{N})}} \,\|\, \mathbb{P}^{\pi,T}_{G^{(\mathbb{N})}})$ reads the retailer's sales data under two possible stories. It takes sales histories that can arise when preferences follow $F^{(\mathbb{N})}$, and evaluates how likely those same observations would be if preferences followed $G^{(\mathbb{N})}$. A small value means that the two paths of preferences generate nearly the same sales histories along the assortments selected by π. On the other hand, a large value indicates that sales generated under $F^{(\mathbb{N})}$ would be hard to reconcile with $G^{(\mathbb{N})}$. Thus, the KL divergence tracks differences in paths of preferences through their footprint in sales data.

Lemma 1. $D_{\mathrm{KL}}(\mathbb{P}^{\pi,T}_{F^{(\mathbb{N})}} \,\|\, \mathbb{P}^{\pi,T}_{G^{(\mathbb{N})}}) = \sum_{t=1}^{T} \mathbb{E}_{\mathbb{P}^{\pi,T}_{F^{(\mathbb{N})}}}\Big[\sum_{i \in S^{\pi,t} \cup \{0\}} p_i(S^{\pi,t}, F^t) \log\big(p_i(S^{\pi,t}, F^t)/p_i(S^{\pi,t}, G^t)\big)\Big]$.

Lemma 1 unpacks this footprint at the level of the assortments offered to each customer. Along histories generated by $F^{(\mathbb{N})}$, the strategy π chooses the assortment $S^{\pi,t}$. For that assortment, the inner sum compares (with the KL divergence) the purchasing distribution induced by F^t with the one induced by G^t, using only the alternatives in $S^{\pi,t} \cup \{0\}$. Products that are not offered do not contribute to distinguishing the two paths in period t. The outer expectation then averages these period-by-period comparisons over the histories generated under $F^{(\mathbb{N})}$. Consequently, two paths differ in this measure only where their purchasing distributions differ on assortments that π actually offers.

4.2 Measuring depreciation along a strategy

For a path $F^{(\mathbb{N})} \in \mathcal{F}$, we compare the sales history generated under $F^{(\mathbb{N})}$ with the sales history generated under its one-period delayed version $F^{(\mathbb{N}-1)}$, which assigns preferences F^{t-1} to customer t. The comparison evaluates how well each customer's preferences describe the preferences of the next incoming customer. Applying Lemma 1 to these two paths of preferences gives:

$$D_{\mathrm{KL}}(\mathbb{P}^{\pi,T}_{F^{(\mathbb{N})}} \,\|\, \mathbb{P}^{\pi,T}_{F^{(\mathbb{N}-1)}}) = \mathbb{E}_{\mathbb{P}^{\pi,T}_{F^{(\mathbb{N})}}}\Big[\sum_{t=2}^{T} \mathcal{K}^t(S^{\pi,t})\Big], \tag{1}$$

where $\mathcal{K}^t(S) := \sum_{i \in S \cup \{0\}} p_i(S, F^t) \log\big(p_i(S, F^t)/p_i(S, F^{t-1})\big)$ is the KL divergence between the purchasing distributions induced by preferences F^t and F^{t-1} on assortment S.

The *sales-information depreciation* in (1) captures the evidence, accumulated along the assortments selected by the strategy, that one customer's preferences may no longer explain those of the next customer. If the resulting divergence is small, then the sales histories generated by $F^{(\mathbb{N})}$ and by its delayed counterpart look similar under those assortments. Such a small divergence can occur because preferences change little, or because the changes are not visible along the selected assortments.

Conversely, a large divergence means that the preferences from one customer poorly explain those from the next one, so the information contained in past sales depreciates. The measure therefore depends on the assortment strategy as changes are counted only when they alter the purchasing distributions induced by assortments actually selected by the retailer. Hence, sales-information depreciation is governed not only by the size of changes in preferences, but also by their *detectability*: whether those changes leave a trace in the sales histories generated by the assortments offered by the retailer.

4.3 Removing the dependence on the assortment strategy

The measure in (1) evaluates sales-information depreciation along the assortments selected by π. Since the retailer specifies a risk profile before choosing how to respond (see Section 3), we also introduce a measure that does not depend on the assortment strategy. We define:

$$\mathcal{M}(F^{(\mathbb{N})}, T) := \sum_{t=2}^{T} \max \left\{ \mathcal{K}^t\left(S\right) : S \in \mathcal{S} \right\},$$

as the *magnitude* of a path of preferences $F^{(\mathbb{N})} \in \mathcal{F}$. Thus, the magnitude measures the maximum sales-information depreciation that could be revealed by a feasible assortment in each period.

Remark 1 (Comparison with existing variation measures)**.** The magnitude is defined along two dimensions: how changes are measured over time, and what is compared. First, it measures changes from one period to the next, in line with non-stationary bandit problems, where variation is measured through changes in expected rewards (Besbes et al. 2019) or in underlying parameters (Cheung et al. 2022). Second, it compares purchasing distributions rather than rewards or parameters, because these distributions determine the sales histories observed by the retailer under each assortment strategy.

The closest existing variation measure in dynamic assortment planning is the one by Foussoul et al. (2023) for MNL preferences. We denote by $\mathcal{F}_{\text{MNL}}$ the set of such paths. For a path $F^{(\mathbb{N})} \in \mathcal{F}_{\text{MNL}}$, let $\nu^t \equiv \nu(F^t) := (\nu_i^t : i \in \mathcal{N})$ be the attraction vector of customer $t \in \mathbb{N}$, so that $p_i(S, F^t) = \nu_i^t(1 + \sum_{j \in S} \nu_j^t)^{-1}$ for $i \in S$. Their measure tracks variation in attraction parameters $(\nu^t : t \in [T])$, whereas our magnitude tracks variation in purchasing distributions. This distinction has two implications.

(i) For paths in $\mathcal{F}_{\text{MNL}}$, their variation measure dominates our magnitude $\mathcal{M}$. Nevertheless, two preference paths may have the same variation in attraction parameters but different magnitudes. This distinction matters for robust planning because paths that appear equivalent under their measure may require different levels of protection and therefore call for different responses under ours.

(ii) In richer choice models, different utility representations may induce the same purchasing distributions. A variation measure defined over utility parameters may then record changes in representation rather than changes in purchasing behavior. For non-parametric choice models (van Ryzin and Vulcano 2015), there may simply be no canonical utility parameterization to track. By working directly with purchasing distributions, our magnitude avoids these model-specific ambiguities.

Finally, in contrast to other measures from the literature, the sales-information depreciation in (1) keeps track of *where* a change is visible. Two preferences paths may change by comparable amounts, yet differ in whether those changes appear under the assortments selected by the retailer. This distinction matters because sales from one assortment might reveal information about changes in preferences that affect other assortments. An assortment strategy can exploit this shared structure to detect changes more effectively than a strategy that treats assortments as independent arms; see Section 5. ■

5 Translating risk profiles into robust planning

We consider two extremes in how preferences may evolve: paths with a single change and paths with an arbitrary number of changes but bounded magnitude. We use the variation measure from Section 4 to turn these regimes into risk profiles that differ in the protection they provide. Each profile specifies the paths of preferences against which the retailer seeks protection and, in turn, a robust-planning response suited to that profile. For each, we quantify the price of robustness and identify a strategy that attain the corresponding regret guarantees; Figure 1 summarizes our contributions.

5.1 Static preferences and beyond

Static preferences. We begin by considering the case in which customers' preferences remain static throughout the selling horizon. This case serves as the reference regime for much of the literature; recall Section 2. For $\gamma_T \geq 0$, define the class of static preferences paths by:

$$\mathcal{F}_S \equiv \mathcal{F}_S(\gamma_T) := \left\{ F^{(\mathbb{N})} \in \mathcal{F} \ : \ \begin{array}{l} F^{t-1} = F^t \quad \forall\, t \geq 2 \quad \text{ and} \\ r(S^*(F^t), F^t) - r(S, F^t) \geq \gamma_T \quad \forall\, t \geq 1, \ \forall\, S \in \mathcal{S} \setminus \{S^*(F^t)\} \end{array} \right\}.$$

The first condition ensures that all customers have the same preferences. The second one separates the revenue-maximizing assortment from all alternatives by requiring a uniform optimality gap. The gap γ_T can be fixed, as in instance-dependent guarantees (Sauré and Zeevi 2013), or decrease with the horizon when the retailer wants adversarial guarantees (Agrawal et al. 2019). For every path $F^{(\mathbb{N})} \in \mathcal{F}_S$, consecutive preferences induce identical purchasing distributions, and therefore $\mathcal{M}(F^{(\mathbb{N})}, T) = 0$.

We use policies $\mathcal{A}$ designed for static preferences as subroutines inside our robust-planning strategies. The retailer uses such a policy to learn the revenue-maximizing assortment from sales data, and its regret under static preferences provides the baseline guarantees. Robustness is measured by the additional regret incurred when this policy is embedded in a strategy π that may restart it, test whether its sales history remains informative, or discard that history once it no longer describes the purchasing behavior of arriving customers. We use $\mathcal{A}$ to denote the downstream policy, and reserve π for the retailer's assortment strategy designed to achieve robustness to evolving customers' preferences.

For a policy $\mathcal{A} \in \mathcal{P}$, we define its adversarial regret under static preferences by:

$$\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, T) := \sup \Big\{ \sum_{t \leq T} r(S^*(F^1), F^1) - \sum_{t \leq T} \mathbb{E}\big[r(S^{\mathcal{A},t}, F^1)\big] \ : \ F^{(\mathbb{N})} \in \mathcal{F}_S \Big\}.$$

This definition lets us state regret guarantees for static preferences without committing to a particular choice model or policy. For MNL preferences, fixed-gap analyses take $\gamma_T \equiv \gamma > 0$ to obtain logarithmic regret, such as $\mathcal{O}(\log^2 T)$ in Rusmevichientong et al. (2010) and $\mathcal{O}(\log T)$ in Sauré and Zeevi (2013). Allowing γ_T to decrease with T instead covers adversarial regimes. The retailer can then rely on the upper-confidence-bound policy of Agrawal et al. (2019) or the Thompson-sampling policies of Agrawal et al. (2026), which achieve $\widetilde{\mathcal{O}}(\sqrt{T})$ regret. For richer choice models, the policy of Li et al. (2025), designed for MCC preferences, achieves $\mathcal{O}(T^{2/3}\log T)$ regret.

Abruptly changing preferences. We consider the simplest departure from static preferences, namely paths that undergo a single abrupt change. This departure captures *high-velocity events*, such as a pandemic, a sudden trend, or a broad economic shock, that change customers' preferences from one regime to another over a short period of time. We define the corresponding class by:

$$\mathcal{F}_A \equiv \mathcal{F}_A(\gamma_T) := \left\{ F^{(\mathbb{N})} \in \mathcal{F} \;:\; \begin{array}{lcl} \exists\, F_1^{(\mathbb{N})}, F_2^{(\mathbb{N})} \in \mathcal{F}_S,\, \tau \geq 2 \text{ s.t. } & & S^*(F_1^1) \neq S^*(F_2^1) \\ F^t = F_1^1 \quad \forall\, t < \tau & \text{and} & F^t = F_2^1 \quad \forall\, t \geq \tau \end{array} \right\}.$$

For a path $F^{(\mathbb{N})} \in \mathcal{F}_A$, the time τ is the *change time*. Before that period, all customers' utilities are drawn from one distribution, and from that period onward they are drawn from another. The first condition ensures that the revenue-maximizing assortment before and after the change are not the same.

Remark 2. If $\tau \geq T+1$, then no change in customers' preferences occurs before the end of the horizon, and $D_{\mathrm{KL}}(\mathbb{P}^{\pi,T}_{F^{(\mathbb{N})}} \,\|\, \mathbb{P}^{\pi,T}_{F^{(\mathbb{N}-1)}}) = 0$. If instead $\tau \leq T$, then the path and its one-period delayed version differ only at the change time. Hence, all sales-information depreciation is concentrated in the period in which customers' preferences move from the old regime to the new one. Using (1) then gives $\min\{\mathcal{K}^\tau(S) : S \in \mathcal{S}\} \leq D_{\mathrm{KL}}(\mathbb{P}^{\pi,T}_{F^{(\mathbb{N})}} \,\|\, \mathbb{P}^{\pi,T}_{F^{(\mathbb{N}-1)}}) \leq \max\{\mathcal{K}^\tau(S) : S \in \mathcal{S}\}$. ■

Remark 2 shows two ways an abrupt change may appear in sales data. The upper bound measures what the most informative assortment can reveal, while the lower bound measures what every assortment reveals. A path in $\mathcal{F}_A$ may have large magnitude yet remain invisible in routine sales if the assortments selected by the retailer do not distinguish the old purchasing distribution from the new one. Such a change is *actively* detectable: revealing it requires targeted assortments. By contrast, when the lower bound is positive, every assortment carries evidence of the change, which is therefore *passively* detectable through routine sales. The risk profiles below formalize this distinction.

5.2 Response to passively detectable changes

We begin with the most favorable risk profile, namely the one in which the change in customers' preferences affects the purchasing distribution induced by every assortment. The retailer can continue to offer the assortments selected by the policy $\mathcal{A}$ while periodically reoffering them to test whether past observations still describe the purchasing behavior of the arriving customers.

5.2.1 Passively detectable changes $\mathcal{F}_D$

Broad shocks may affect customers' preferences across the retailer's entire catalog rather than within a particular category or brand. We capture such changes by requiring the purchasing distributions before and after the change to differ under every assortment. Formally, we define the risk profile:

$$\mathcal{F}_D \equiv \mathcal{F}_D(\gamma_T, \varepsilon) := \{F^{(\mathbb{N})} \in \mathcal{F}_A \,:\, \exists\, \tau \geq 2 \text{ s.t. } \min\{\mathcal{K}^\tau(S) \,:\, S \in \mathcal{S}\} \geq \varepsilon\}.$$

The profile $\mathcal{F}_D$ contains abrupt changes for which the purchasing distributions before and after the change differ by a sufficiently large amount under every assortment. The parameter ε measures how visible the change is in routine sales data. If the retailer restricts attention to MNL preferences, then paths in $\mathcal{F}_D$ are those for which every product's attraction parameter differs across the two regimes.

5.2.2 Price of robustness

Before proposing a robust-planning strategy, we first identify the price of robustness for the risk profile $\mathcal{F}_D$. For paths of preferences in $\mathcal{F}_D$, the change affects the purchasing distribution induced by every feasible assortment, but its timing still remains unknown to the retailer. Any strategy must therefore distinguish a genuine change in preferences from random variation in customers' choices.

Proposition 1. *There exists $c > 0$ and $T_0 > 0$ such that for all $T \geq T_0$:*

$$\mathcal{R}^*\left(\mathcal{F}_D, T\right) \geq c \log T.$$

Proposition 1 shows that robustness to $\mathcal{F}_D$ has an unavoidable cost. To see why, consider a path of preferences with change time τ, and denote by $S^*(F^1)$ and $S^*(F^\tau)$ the revenue-maximizing assortments before and after the change, respectively. A strategy with small regret for every possible change time must behave in two opposite ways. Before τ, it cannot offer $S^*(F^\tau)$ too often, because that assortment is suboptimal. After τ, it cannot keep offering $S^*(F^1)$ for too long, because that assortment is suboptimal. Hence, low regret forces the strategy to move from $S^*(F^1)$ to $S^*(F^\tau)$ soon after the change. The proof turns this implication into a stopping procedure: it observes the assortments selected by the strategy and declares that the strategy has identified the change once $S^*(F^\tau)$ has been offered often enough. However, no stopping procedure using only the information available to the retailer can uniformly locate the change time within fewer than $\mathcal{O}(\log T)$ customer interactions.

Specifically, for any change time, with probability bounded away from zero, the stopping procedure needs order $\mathcal{O}(\log T)$ customer interactions before or after the change to identify it. If it reacts too late, then the strategy keeps selecting $S^*(F^1)$ after the change; if it reacts too early, then the strategy moves toward $S^*(F^\tau)$ while the old regime is still in place. Either way, the strategy offers a suboptimal assortment for $\mathcal{O}(\log T)$ periods, which gives the logarithmic lower bound.

5.2.3 Robustness against passively detectable changes

We now show that robustness can be achieved efficiently by adding a monitoring layer to the downstream policy $\mathcal{A}$. The retailer follows the assortments recommended by $\mathcal{A}$, and uses the sales generated by these assortments to test whether preferences have changed. The test is agnostic to the choice model characterizing customers' preferences: it compares two sales histories generated under the same assortment, and the strategy restarts $\mathcal{A}$ when their empirical purchasing distributions differ enough.

Algorithm 1 Passive monitoring strategy $\pi(\mathcal{A}, T)$

Input. Policy $\mathcal{A}$ and horizon T

Set $q_T \leftarrow \lceil \log(T) \rceil \lceil \log\log(T) \rceil$, $m_T \leftarrow 8q_T$, $t \leftarrow 0$, $k \leftarrow 1$, and DETECTED $\leftarrow$ False

$(S_0, t) \leftarrow \text{Run}(\mathcal{A}, m_T, t)$, $(\mathcal{J}_0, t) \leftarrow \text{Offer}(S_0, q_T, t)$, and refresh $\mathcal{A}$

while $t < T$ and DETECTED = False **do**

    $(S_k, t) \leftarrow \text{Run}(\mathcal{A}, m_T, t)$ and $(\mathcal{J}_k, t) \leftarrow \text{Offer}(S_k, q_T, t)$

    $(\mathcal{I}_k, t) \leftarrow \text{Offer}(S_{k-1}, q_T, t)$ and DETECTED $\leftarrow \text{Test}(S_{k-1}, \mathcal{J}_{k-1}, \mathcal{I}_k)$ and $k \leftarrow k + 1$

Run a fresh copy of $\mathcal{A}$ on the remaining customers

$\text{Run}(\mathcal{A}, m, t)$ runs the current copy of $\mathcal{A}$ for a block of m customers, updates $\mathcal{A}$ and increases t after each customer, samples uniformly one assortment from the sequence offered during the block, and returns this sampled assortment

$\text{Offer}(S, q, t)$ offers the same assortment S to the next q customers, increases t and returns the resulting sales history

$\text{Test}(S, \mathcal{J}, \mathcal{I})$ is a two-sample test with categorical data from $\mathcal{J}$ and $\mathcal{I}$ for assortment S, and returns True upon rejection

Algorithm 1 turns a two-sample test into a monitoring layer for $\mathcal{A}$. The strategy works in cycles; in each cycle, it first lets $\mathcal{A}$ learn, then selects uniformly at random one assortment previously chosen by $\mathcal{A}$ and offers it repeatedly to create a reference history. During the next cycle, the strategy offers that assortment again and compares the new purchase outcomes with the reference history. If the test rejects equality of the purchasing distributions between the two cycles, then the strategy interprets the discrepancy as evidence of a change in customers' preferences, discards past sales, and restarts $\mathcal{A}$.

When preferences are static and $\mathcal{A}$ has low regret, it eventually offers the revenue-maximizing assortment with high probability. The assortment selected uniformly at random for monitoring is then also likely to be maximizing the revenue, so monitoring imposes essentially no cost on the retailer.

Theorem 1. *There exists $T_0 > 0$ such that for all $T \geq T_0$:*

$$\mathcal{R}^{\pi(\mathcal{A},T)}(\mathcal{F}_D, T) \leq 3\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, T) + 89\|w\|_\infty \log(T) \log\log(T).$$

Theorem 1 shows that passive monitoring makes $\mathcal{A}$ robust to $\mathcal{F}_D$. The adversarial regret remains, up to logarithmic factors, of the same order as the regret of $\mathcal{A}$ under static preferences, namely $\widetilde{\mathcal{O}}(\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, T))$. The retailer can therefore keep using $\mathcal{A}$ to learn the revenue-maximizing assortment, monitor the sales generated by the assortments selected by $\mathcal{A}$, and restart only when the two-sample test detects a change in customers' preferences. When $\mathcal{A}$ has logarithmic regret under static preferences, robustness to passively detectable changes adds only logarithmic factors to the regret.

5.3 Response to actively detectable changes

We turn to the risk profile $\mathcal{F}_U$, where the change is abrupt but not visible under every assortment. Some assortments may generate the same purchasing distribution before and after the change, even though other assortments would reveal it. Robustness cannot rely only on monitoring the assortments offered by $\mathcal{A}$. The retailer must also offer assortments that can reveal the change.

5.3.1 Actively detectable changes $\mathcal{F}_U$

Actively detectable changes can arise when customers' preferences shift within a product category, toward a specific brand, or along a particular product attribute. Assortments that do not expose the affected products may then induce the same purchasing distribution before and after the change. A strategy that monitors only the assortments selected by $\mathcal{A}$ (as discussed in Section 5.2) may then keep collecting sales histories that do not reveal the change. Robustness requires the retailer to offer targeted assortments capable of revealing changes that might be hidden under those selected by $\mathcal{A}$. We capture such changes through the following risk profile:

$$\mathcal{F}_U \equiv \mathcal{F}_U(\gamma_T, \kappa, \phi) := \left\{ F^{(\mathbb{N})} \in \mathcal{F}_A \ : \ \begin{array}{c} \exists\, \tau \geq 2 \text{ s.t. } \min\{\mathcal{K}^\tau(S) : S \in \mathcal{S}\} = 0 \\ \text{and } \max\{\mathcal{K}^\tau(S) : S \in \mathcal{S}\} \in (\kappa, \phi) \end{array} \right\}.$$

The lower bound κ in $\mathcal{F}_U$ ensures that the change in customers' preferences alters the purchasing distribution for at least one feasible assortment, whereas the upper bound ϕ controls the magnitude of the path. The risk profile captures paths of customers' preferences for which the change is detectable under sufficiently informative assortments but may not be visible under others.

5.3.2 Price of robustness

Because some assortments may be uninformative, any robust strategy must spend enough periods offering assortments that can distinguish whether preferences have changed. The resulting price of robustness is no longer logarithmic, as in Proposition 1, but of square-root order, as shown in Proposition 2. This increase reflects the additional experimentation required to make the change visible.

Proposition 2. *There exists $c > 0$ such that for all $T \geq 9$:*

$$\mathcal{R}^*(\mathcal{F}_U, T) \geq c\sqrt{T}.$$

The proof of Proposition 2 divides the horizon into intervals whose endpoints serve as candidate change times. For each pair of consecutive candidates, it compares two paths of preferences: one in which the change occurs at the earlier time and another in which it occurs at the later time. Two possibilities arise: (i) if at least one pair induces similar sales-history distributions, then the strategy cannot reliably determine which change time occurred; (ii) if every pair is well separated, then the strategy must have generated that separation through costly experimentation before the change.

(i) Suppose that some pair of neighboring candidate change times induces sales-history distributions with small KL divergence. The strategy then cannot reliably determine which path of customers' preferences it faces. Under the path with the earlier change, it must move toward the new revenue-maximizing assortment; under the path with the later change, it must continue selecting the old one. The assortments selected between the two candidate times induce a statistical test between these paths. Small KL divergence makes the test unreliable, so the strategy must incur regret on at least one path of customers' preferences because it cannot respond correctly to both.

(ii) Next, suppose that every pair of neighboring candidate change times is well separated in KL divergence. The strategy must then have generated enough evidence to distinguish all such pairs. The proof makes that evidence costly by letting the environment place the change at the end of the horizon and choose a path of customers' preferences for which the revenue-maximizing assortment before the change is uninformative. Distinguishing the candidate change times thus requires offering suboptimal assortments before the change, which directly increases regret.

5.3.3 Robustness against actively detectable changes

We augment the downstream policy $\mathcal{A}$ with a *sentinel* family $\mathcal{E} \subseteq \mathcal{S}$. The family $\mathcal{E}$ contains at least one assortment that can distinguish the purchasing distributions before and after the change. The retailer uses $\mathcal{A}$ to learn the revenue-maximizing assortment, while periodically selecting assortments from $\mathcal{E}$ and applying the statistical test discussed in Section 5.2. These assortments create sales histories that can reveal changes which may remain hidden along the assortments selected by $\mathcal{A}$.

Algorithm 2 Active monitoring strategy $\pi(\mathcal{A}, \mathcal{E}, T)$

Input. Policy $\mathcal{A}$, assortment family $\mathcal{E}$, and horizon T
Set $q_T \leftarrow \lceil \log(T) \log\log(T) \rceil$, $m_T \leftarrow \lceil \sqrt{T|\mathcal{E}|q_T} \rceil$, $t \leftarrow 0$, and Detected $\leftarrow$ False
For each $S \in \mathcal{E}$, set $(\mathcal{J}(S), t) \leftarrow \text{Offer}(S, q_T, t)$
while $t < T$ and Detected $=$ False **do**
    $t \leftarrow \text{Run}(\mathcal{A}, m_T, t)$
    For each $S \in \mathcal{E}$, **while** Detected $=$ False:
        $(\mathcal{I}(S), t) \leftarrow \text{Offer}(S, q_T, t)$ and Detected $\leftarrow \text{Test}(S, \mathcal{J}(S), \mathcal{I}(S))$
Run a new copy of $\mathcal{A}$ on the remaining customers

Run$(\mathcal{A}, m, t)$ runs the current copy of $\mathcal{A}$ for m customers, updates $\mathcal{A}$ after each customer, increases t, and returns t

Offer(S, q, t) offers assortment S to the next q customers, increases t and returns the updated value of t

Test$(S, \mathcal{J}, \mathcal{I})$ is a two-sample test with categorical data from $\mathcal{J}$ and $\mathcal{I}$ for assortment S, and returns True upon rejection

Algorithm 2 implements this idea by interleaving long learning phases with shorter sentinel phases. During learning phases, $\mathcal{A}$ selects assortments. During sentinel phases, each assortment in $\mathcal{E}$ is offered, and the resulting sales are compared with the sales initially collected to form its reference history. If the test rejects equality of the purchasing distributions, then the strategy interprets the discrepancy as evidence of a change in customers' preferences, discards the previous sales history, and restarts $\mathcal{A}$.

The robustness guarantee of Algorithm 2 is tied to the sentinel family $\mathcal{E}$ it receives as input. The strategy can detect a change only through sales histories generated by assortments in $\mathcal{E}$. Hence, its adversarial regret guarantee is meaningful for paths in $\mathcal{F}_U$ whose abrupt change is visible through at least one assortment from $\mathcal{E}$. For $\mathcal{E} \subseteq \mathcal{S}$, define $\eta(\mathcal{E}) := \inf \{ \max\{\mathcal{K}^t(S) \,:\, S \in \mathcal{E},\, t \geq 2\} : F^{(\mathbb{N})} \in \mathcal{F}_U\}$.

The quantity $\eta(\mathcal{E})$ measures how well the sentinel family covers the changes in $\mathcal{F}_U$. Thus, $\eta(\mathcal{E}) > 0$ means that every path in $\mathcal{F}_U$ produces a change that can be detected by offering at least one assortment in $\mathcal{E}$. Larger values make detection easier, whereas values close to zero mean that some paths are difficult to distinguish using only sales generated by assortments from $\mathcal{E}$.

Remark 3. For MNL preferences, a family of assortments that covers all products is sufficient to make every change in $\mathcal{F}_U$ visible. Indeed, each product $i \in \mathcal{N}$ then appears in at least one sentinel assortment $S \in \mathcal{E}$, and a change in its attractiveness affects every assortment containing it. Hence, if the assortments in $\mathcal{E}$ cover all products, then $\eta(\mathcal{E}) > 0$. Extending the result beyond MNL preferences is less immediate because richer substitution patterns may make visibility depend on specific product combinations rather than on product coverage alone. ■

Theorem 2. *If $\eta(\mathcal{E}) > 0$, then there exists $T_0 > 0$ such that for all $T \geq T_0$:*

$$\mathcal{R}^{\pi(\mathcal{A},\mathcal{E},T)}(\mathcal{F}_U, T) \leq 2\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, T) + 9\|w\|_\infty \sqrt{T|\mathcal{E}| \log(T) \log\log(T)}.$$

Theorem 2 shows that robustness to $\mathcal{F}_U$ can be obtained by adding a monitoring layer to $\mathcal{A}$. The regret has two components: the regret inherited from $\mathcal{A}$ under static preferences and the monitoring cost, which scales with the size of the sentinel family $|\mathcal{E}|$. The effectiveness of this monitoring layer depends on whether $\mathcal{E}$ can reveal changes, as captured by the condition $\eta(\mathcal{E}) > 0$. This condition holds, for instance, for MNL preferences under the product-cover construction of Remark 3, in which case $|\mathcal{E}| = \mathcal{O}(N/K)$. When $\mathcal{A}$ satisfies $\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, T) = \widetilde{\mathcal{O}}(\sqrt{T})$, the active monitoring strategy matches the lower bound in Proposition 2 up to logarithmic factors, attaining regret of order $\widetilde{\mathcal{O}}(\sqrt{T|\mathcal{E}|})$.

5.4 Response to sales-information depreciation

We now consider the risk profile $\mathcal{F}_G$, where paths of customers' preferences may evolve arbitrarily over the horizon subject only to a budget constraint on their magnitude. Considering that robust planning can be achieved in multiple ways, we focus on one response based on periodic restarts. In that response, the key decision is how long $\mathcal{A}$ can keep using its sales history before discarding it completely.

5.4.1 General sales-history depreciation $\mathcal{F}_G$

The risk profile $\mathcal{F}_G$ constrains the path of preferences only through a magnitude budget M_T. Unlike $\mathcal{F}_D$ and $\mathcal{F}_U$, it is not organized around a single change time. It does not specify how preferences move, whether that movement is detectable from routine sales, or which assortments reveal it.

Instead, $\mathcal{F}_G$ records how much the information contained in sales histories may depreciate over the horizon. The budget M_T is then used to control this depreciation. Formally, define the risk profile:

$$\mathcal{F}_G \equiv \mathcal{F}_G(\gamma_T, M_T) := \left\{ F^{(\mathbb{N})} \in \mathcal{F} \, : \, \mathcal{M}(F^{(\mathbb{N})}, T) \leq M_T \text{ and } (F^s \, : \, t \in \mathbb{N}) \in \mathcal{F}_S(\gamma_T) \; \forall \, s \in \mathbb{N} \right\}.$$

This risk profile contains the profiles studied before as special cases. Static preferences have zero magnitude, while abrupt changes concentrate magnitude at the change time. The risk profile $\mathcal{F}_G$ subsumes the preceding risk profiles while also allowing the depreciation of sales information to be spread across the horizon rather than concentrated in a single period. In that sense, it supports protection against a broader range of preference paths than the risk profiles considered earlier.

5.4.2 Price of robustness

The price of robustness under $\mathcal{F}_G$ is governed by how long the strategy retains its sales history. A longer history gives the strategy more observations to identify the revenue-maximizing assortment when preferences are nearly static, but also allows sales generated under earlier purchasing behavior to influence decisions for longer. A shorter history, on the other hand, limits that influence at the cost of repeatedly relearning the revenue-maximizing assortment. The magnitude budget M_T thus determines how the retailer balances relearning costs against sales-history depreciation.

Proposition 3. *Assume that $w_i = 1$ for all $i \in \mathcal{N}$, $1 \leq M_T \leq T/8$ and $3 \leq K + 1 \leq N$. There exist $c_1 > 0$ and $\tilde{\mathcal{F}} \subseteq \mathcal{F}$ with $\sup\{\mathcal{M}(F^{(\mathbb{N})}, T) \, : \, F^{(\mathbb{N})} \in \tilde{\mathcal{F}}\} \leq 4M_T/3$ and:*

$$\mathcal{R}^*(\tilde{\mathcal{F}}, T) \geq c_1 T^{5/6} M_T^{1/6}.$$

The proof of Proposition 3 constructs paths of MNL preferences that are particularly challenging for any strategy. The horizon is divided into intervals. In each interval, the retailer must distinguish between two nearly tied assortments, one of which has a revenue advantage that grows and then fades gradually. Across intervals, the revenue-maximizing assortment changes. The proof converts the assortments selected by the strategy inside an interval into a statistical test to identify the revenue-maximizing assortment. Since the two induced sales-history distributions are close, no test can identify the revenue-maximizing assortment with sufficiently high probability. The retailer must either devote enough periods to distinguishing the two assortments or incur regret by selecting the wrong one.

5.4.3 Robustness without assumptions on the subroutine

Without additional structure on $\mathcal{A}$, robustness to $\mathcal{F}_G$ can be obtained by controlling how old is the sales history given to the downstream policy. We partition the horizon into intervals of length Δ and restart $\mathcal{A}$ at the beginning of each interval. At the beginning of each interval, the policy starts with an empty history; at the end of the interval, the accumulated observations are discarded. The parameter Δ determines how much past sales information the retailer is willing to reuse.

Algorithm 3 Periodic restarting strategy $\pi(\mathcal{A}, T, \Delta)$

Input: Policy $\mathcal{A}$, horizon T, and batch-size Δ

while $1 \le j \le \lceil T/\Delta \rceil$ **do**

  Run a new copy of $\mathcal{A}$ on customer $t = (j-1)\Delta + 1$ to $t = \min\{j\,\Delta, T\}$ and then $j = j+1$

Algorithm 3 summarizes this strategy. The guarantee below treats $\mathcal{A}$ as completely arbitrary. It uses only its regret under static preferences over an interval of length Δ. In particular, the analysis does not require any assumption on how $\mathcal{A}$ reacts to mild changes in preferences inside an interval.

Theorem 3. *There exists $c > 0$ such that for all $T \ge 2$:*

$$\mathcal{R}^{\pi(\mathcal{A},T,\Delta)}(\mathcal{F}_G, T) \le \lceil T/\Delta \rceil \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, \Delta) + c\|w\|_\infty \Delta^{3/2}\sqrt{2TM_T}.$$

The first term in Theorem 3 is the regret that $\mathcal{A}$ would achieve over the $\lceil T/\Delta \rceil$ intervals if preferences were fixed within each interval. The second term accounts for sales-history depreciation: within an interval, $\mathcal{A}$ may use observations collected many periods earlier, whose relevance can deteriorate as preferences change. Increasing Δ reduces the number of restarts, but increases the age of the observations used by $\mathcal{A}$. A smaller Δ restarts the policy more frequently, but gives $\mathcal{A}$ fewer observations in each interval to identify the revenue-maximizing assortment.

Corollary 1. *Assume that $\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, n) \le C_{\mathcal{A}}\sqrt{n}$. Let $C = \sqrt{2}c\|w\|_\infty$ with $c > 0$ as in Theorem 3, and choose $\Delta = \lceil (2C_{\mathcal{A}}/3C)^{1/2}(T/M_T)^{1/4} \rceil$. If T is large enough so that $\Delta \in [T]$, then:*

$$\mathcal{R}^{\pi(\mathcal{A},T,\Delta)}(\mathcal{F}_G, T) \le 7C_{\mathcal{A}}^{3/4}C^{1/4}T^{7/8}M_T^{1/8}.$$

Corollary 1 optimizes the regret bound from Theorem 3 over restart frequencies when the downstream policy $\mathcal{A}$ achieves $\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, n) = \mathcal{O}(\sqrt{n})$ under static preferences. The resulting guarantee does not match the lower bound in Proposition 3 because the analysis treats $\mathcal{A}$ as arbitrary. On each interval, it bounds the regret of $\mathcal{A}$ by the regret it would incur under static preferences plus a worst-case sales-history depreciation term, which is charged even when preferences change only slightly within the interval. The bound is therefore conservative: it must allow poor assortment decisions even when the revenue information contained in recent sales has changed only slightly. The next subsection strengthens the analysis by requiring $\mathcal{A}$ to remain reliable on such intervals.

5.4.4 Improved rates with a near-stationary robust subroutine

The previous guarantee treats $\mathcal{A}$ as arbitrary and uses its performance only under static preferences. Such a guarantee is deliberately conservative. It ignores useful structure: if the revenue generated by each assortment changes only mildly, then observations collected early in the interval may still help the policy compare assortments later in the interval. To formalize this idea, we measure change in revenue inside an interval. For an interval $I = \{t_1, \dots, t_n\} \subseteq [T]$ and a path $F^{(\mathbb{N})} \in \mathcal{F}$, define $\mathcal{V}_I(F^{(\mathbb{N})}) := \sum_{j=2}^{n} \max\{|r(S, F^{t_j}) - r(S, F^{t_j - 1})| : S \in \mathcal{S}\}$. The quantity $\mathcal{V}_I(F^{(\mathbb{N})})$ is small when the revenue of every assortment changes only mildly from one period to the next.

Definition 1. *We say that $\mathcal{A} \in \mathcal{P}$ is near-stationary robust for $\mathcal{F}$ if there exists $c_{\mathcal{A}} > 0$ such that, for every interval $I \subseteq [T]$ and every path $F^{(\mathbb{N})} \in \mathcal{F}$, when $\mathcal{A}$ is started at the beginning of I, one has:*

$$\max\{\sum_{t\in I} r(S, F^t) \,:\, S \in \mathcal{S}\} - \mathbb{E}[\sum_{t\in I} r(S^{\mathcal{A},t}, F^t)] \le c_{\mathcal{A}}\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, |I|) + c_{\mathcal{A}}|I|\mathcal{V}_I(F^{(\mathbb{N})}).$$

The condition in Definition 1 ties the regret of $\mathcal{A}$ on any interval I, measured relative to the best fixed assortment in hindsight over I, to two terms: its regret under static preferences, as defined in Section 5.1, and the revenue variability $|I|\mathcal{V}_I(F^{(\mathbb{N})})$ within the interval. To be precise, the left-hand side of the condition is the regret of $\mathcal{A}$ relative to an oracle that knows the preferences $(F^t : t \in I)$ but can choose only a single assortment for the whole interval to maximize total expected revenue. If preferences are static on I, then $\mathcal{V}_I(F^{(\mathbb{N})}) = 0$, and the condition reduces to the usual guarantee for static preferences. When preferences evolve, the additional term $|I|\mathcal{V}_I(F^{(\mathbb{N})})$ separates the learning cost from the fact that the revenue generated by a fixed assortment may change over the interval.

In other words, even if the policy has identified a good assortment early in the interval, that same assortment may generate different revenues later. The discrepancy is not reflected in the regret term $\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, |I|)$. Thus, near-stationary robustness captures the idea that $\mathcal{A}$ remains reliable relative to the revenue-maximizing assortment over the interval when preferences move little in revenue terms.

Theorem 4. *If $\mathcal{A}$ is near-stationary robust for $\mathcal{F}_G$, then for all $T \ge 1$:*

$$\mathcal{R}^{\pi(\mathcal{A},T,\Delta)}(\mathcal{F}_G, T) \le c_{\mathcal{A}}\lceil T/\Delta\rceil\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, \Delta) + (2 + c_{\mathcal{A}})\sqrt{2}\|w\|_\infty\Delta\sqrt{TM_T}.$$

Theorem 4 keeps the regret term associated with $\mathcal{A}$ unchanged relative to Theorem 3, but gives sharper control of the sales-information depreciation term. The proof partitions the horizon into the intervals induced by the restarts. On each interval, near-stationary robustness bounds the regret of $\mathcal{A}$ by its regret under static preferences plus the local revenue variation $\mathcal{V}_I(F^{(\mathbb{N})})$. Summing over intervals gives the first term, and the accumulated local revenue variation is controlled through the magnitude budget M_T. Hence, the analysis better controls the revenue changes that occur within intervals, rather than imposing a conservative bound merely because preferences may vary within the interval.

Corollary 2. *Assume that $\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, n) \le C_{\mathcal{A}}\sqrt{n}$. Choose $\Delta = \lceil (C_{\mathcal{A}}^2 T/M_T)^{1/3}\rceil$. If T is large enough so that $\Delta \in [T]$, then there exists $c > 0$ such that:*

$$\mathcal{R}^{\pi(\mathcal{A},T,\Delta)}(\mathcal{F}_G, T) \le c(C_{\mathcal{A}}^{2/3}T^{5/6}M_T^{1/6}).$$

Corollary 2 matches the lower bound from Proposition 3 up to constants. The restart length $\Delta = \lceil (C_{\mathcal{A}}^2 T/M_T)^{1/3}\rceil$ reflects the role of the magnitude budget in the robust-planning strategy. When sales information depreciates more (i.e., high magnitude), the retailer restarts more frequently; when older observations remain useful for longer (i.e., low magnitude), the retailer use lower restart frequencies.

Remark 4. The policy of Foussoul et al. (2023) is designed for MNL preferences, and their analysis establishes that it continues to learn effectively when attraction parameters vary only mildly over time. Because variation in these parameters controls the induced purchasing distributions and the resulting expected revenues, their results imply that the policy is near-stationary robust on $\mathcal{F}_{\mathrm{MNL}}$. ■

Remark 4 identifies a case in which the periodic restarting strategy in Algorithm 3 can use an existing policy to achieve the regret lower bound up to logarithmic factors. In general, however, achieving robustness to the risk profile $\mathcal{F}_G$ is challenging. The retailer must determine its risk appetite by choosing the magnitude budget M_T, which specifies how much the information contained in past sales may depreciate over the horizon, and assess whether the downstream policy remains reliable when revenues vary mildly within a restart interval. Existing guarantees typically provide regret bounds under static preferences but do not quantify performance under such mild changes. Consequently, we next study a second regime in which the retailer does not specify a risk profile in advance.

6 Robust planning under an unspecified risk profile

The previous section gives a simple way to turn a policy designed for static preferences into a robust-planning strategy: before the selling season, the retailer specifies a risk profile. Quantifying the corresponding price of robustness then reveals how the strategy can fail on paths of preferences in that profile, and thereby points to a response sufficient to protect against those paths. The construction is appealing because it makes the robustness guarantee and the corresponding response transparent. Its limitation is equally clear. The guarantee is inherited from the risk profile used to design the robust-planning strategy, and therefore may not extend to paths of preferences outside that profile.

Next, we study the situation in which this limitation becomes binding. The retailer still seeks protection against evolving customers' preferences but cannot specify a risk profile before the selling season. In that case, we adopt a conservative perspective. Over a finite horizon, every path of preferences has a well-defined magnitude and can thus be viewed as belonging to $\mathcal{F}_G$ for a suitable parameter value. The retailer, however, does not know this value in advance and hence cannot determine the appropriate magnitude budget governing the restart frequencies prescribed in Corollaries 1 and 2.

Algorithm 4 Learning to restart strategy $\pi(\mathcal{A}, T)$

Input. Policy $\mathcal{A}$ and horizon T

$b_T \leftarrow \lceil T^{2/3} \rceil$, $v^1(\Delta) \leftarrow 1$ for every $\Delta \in \mathcal{D}_T := \{2^\ell : 0 \le \ell \le \lfloor \log_2 b_T \rfloor\}$

for $1 \le k \le \lfloor T/b_T \rfloor$ **do**

  $q^k \leftarrow v^k / \|v^k\|_1$, $\Delta^k \sim q^k$, and $v^{k+1} \leftarrow v^k$

  $r^k \leftarrow \mathrm{Run}(\mathcal{A}, b_T, \Delta^k)$ and $v^{k+1}(\Delta^k) \leftarrow \mathrm{Update}(v^k, r^k, \Delta^k)$

Run a new copy of $\mathcal{A}$ on the remaining customers

$\mathrm{Run}(\mathcal{A}, b_T, \Delta^k)$ runs $\pi(\mathcal{A}, b_T, \Delta^k)$, collects the achieved revenue $\tilde{r}^k$ over that cycle and returns $r^k := 1 - \tilde{r}^k / (\|w\|_\infty b_T)$

$\mathrm{Update}(v^k, r^k, \Delta^k) = v^k(\Delta^k) \exp(-\eta_T r^k (q^k(\Delta^k) + \eta_T/2)^{-1})$, for $\eta_T := \min\{1, (\log\log(b_T))^{1/2} (\lfloor T/b_T \rfloor \log(b_T))^{-1/2}\}$

Algorithm 4 addresses this uncertainty by adding a learning layer to the periodic restarting strategy of Algorithm 3. Unlike Algorithm 3 in which the magnitude budget fixes the restart frequency (e.g., by using Corollaries 1 and 2), Algorithm 4 keeps several restart frequencies available and uses the revenue generated throughout the selling season to update how likely each frequency is to be selected. Hence, the restart frequency is not fixed in advance, but learned from the sales collected.

The strategy follows a *bandit-over-bandit* construction (Agarwal et al. 2017; Besbes et al. 2019; Cheung et al. 2022). Each arm is a periodic restarting strategy that runs its own copy of the downstream policy $\mathcal{A}$ at a different restart frequency. In each interval, Algorithm 4 selects one arm, runs the corresponding strategy, and uses the adversarial bandit algorithm of Neu (2015) to increase the probability of selecting frequencies that generate larger revenue. Consequently, the retailer learns at two levels: each copy of $\mathcal{A}$ learns the revenue-maximizing assortment from the sales collected since its most recent restart, while the outer level learns which restart frequency performs best across intervals. This additional experimentation is precisely the operational price of leaving the risk profile unspecified.

6.1 Robustness without assumptions on the subroutine

We first analyze Algorithm 4 without assuming near-stationary robustness of $\mathcal{A}$. As in Section 5.4.3, we use only its regret guarantee under static preferences and otherwise leave its behavior unrestricted. The following result quantifies the cost of learning the restart frequency in this general case.

Theorem 5. *There exists $c_1 > 0$ such that for all $T \geq 8$:*

$$\mathcal{R}^{\pi(\mathcal{A},T)}(\mathcal{F}_G, T) \leq c_1\big(\min\big\{\frac{T}{\Delta}\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, \Delta) + \Delta^{3/2}\sqrt{TM_T} \,:\, \Delta \in \mathcal{D}_T\big\} + T^{5/6}\log(T)\big).$$

Theorem 5 characterizes the price of not specifying a risk profile before the selling season. If M_T were known, then the retailer could choose a restart frequency that balances two costs. Frequent restarts prevent old sales histories from guiding decisions for too long, but force $\mathcal{A}$ to relearn on shorter horizons. Infrequent restarts expose $\mathcal{A}$ to the risk of evolving preferences. The minimum term in Theorem 5 is the optimal frequency that could have been chosen with knowledge of the magnitude (Theorem 3). Our strategy replaces knowledge of the magnitude with experimentation over restart frequencies, and the additional term in the regret bound is the cost of that experimentation.

Corollary 3. *Assume that $\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, n) \leq C_{\mathcal{A}}\sqrt{n}$. There exists $c_2 > 0$ such that for all $T \geq 8$:*

$$\mathcal{R}^{\pi(\mathcal{A},T)}(\mathcal{F}_G, T) \leq c_2 \begin{cases} C_{\mathcal{A}}^{3/4}T^{7/8}M_T^{1/8} + T^{5/6}\log T, & \text{if } M_T \geq C_{\mathcal{A}}^2/T^{5/3}, \\ C_{\mathcal{A}}T^{2/3} + T^{5/6}\log T, & \text{if } M_T < C_{\mathcal{A}}^2/T^{5/3}. \end{cases}$$

Corollary 3 separates two robustness regimes. When the unknown magnitude M_T is large enough, the policy matches the guarantee from Corollary 1 in which the magnitude is known, up to the additional cost of learning the restart frequency. On the other hand, when M_T is small, preferences move too little for frequent restarts to be necessary, and the lowest restart frequency controls the guarantee. In that regime, the main revenue loss comes from experimenting with higher restart frequencies that, in hindsight, would not have been selected with knowledge of M_T.

6.2 Improved rates with a near-stationary subroutine

The previous section obtains regret guarantees by using only the performance of downstream policy $\mathcal{A}$ under static preferences and imposing no additional robustness requirement. We now show that the guarantee improves when $\mathcal{A}$ satisfies the near-stationary robustness condition in Definition 1.

Theorem 6. *If $\mathcal{A}$ is near-stationary robust for $\mathcal{F}_G$, then there exists $c_1 > 0$ such that for all $T \geq 8$:*

$$\mathcal{R}^{\pi(\mathcal{A},T)}(\mathcal{F}_G, T) \leq c_1\big(\min\{\frac{T}{\Delta}\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, \Delta) + \Delta\sqrt{TM_T} \,:\, \Delta \in \mathcal{D}_T\} + T^{5/6}\log(T)\big).$$

Theorem 6 shows how near-stationary robustness affects the performance guarantees of Algorithm 4. The first term in the regret bound recovers the guarantee from Theorem 4 for the best restart frequency in hindsight. The second term is the cost of not knowing the magnitude in advance: the policy must experiment across restart frequencies before finding the appropriate one.

Corollary 4. *Assume that $\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S, n) \leq C_{\mathcal{A}}\sqrt{n}$. There exists $c_2 > 0$ such that for all $T \geq 8$:*

$$\mathcal{R}^{\pi(\mathcal{A},T)}(\mathcal{F}_G, T) \leq c_2 \begin{cases} C_{\mathcal{A}}^{2/3} T^{5/6} M_T^{1/6} + T^{5/6}\log T, & \text{if } M_T \geq C_{\mathcal{A}}^2/T, \\ C_{\mathcal{A}} T^{2/3} + T^{5/6}\log T, & \text{if } M_T < C_{\mathcal{A}}^2/T. \end{cases}$$

Corollary 4 shows that near-stationary robustness recovers the main dependence obtained when the magnitude is known. When the magnitude is above the threshold, regret scales as $\widetilde{\mathcal{O}}(T^{5/6}M_T^{1/6})$, matching the dependence in Corollary 2 up to the additional experimentation cost caused by not knowing the magnitude. Thus, when the downstream policy is near-stationary robust, as is the policy of Foussoul et al. (2023) for MNL preferences (Remark 4), the retailer can attain the same robustness guarantee as if it had planned with the appropriate magnitude budget. When the magnitude is below the threshold, however, preferences move too little to justify frequent restarts, and the guarantee is driven by experimentation over restart frequencies that would have been unnecessary in hindsight with knowledge of that magnitude.

These results indicate that robustness to evolving preferences is possible without specifying a risk profile in advance. That flexibility, however, has a price: the retailer must experiment with restart frequencies to learn how to balance the remembering-forgetting trade-off. Leaving the risk profile unspecified might reflect two different concerns. If the retailer is concerned that a chosen magnitude budget may be too small to protect against the evolving preferences it will face, then experimentation across restart frequencies is a price to pay for robustness, and Algorithm 4 provides a way to pay that price. If, instead, the retailer worries that a conservative risk profile would lead to unnecessary restarts when preferences move little, then the strategy can be refined; see Remark 5.

Remark 5. The experimentation in Algorithm 4 can be overly conservative when preferences move little as the strategy may try high restart frequencies even though lower ones would have been sufficient. A natural refinement is to combine the strategy with monitoring similar to that used in Algorithm 2, so that experimentation over restart frequencies begins only when sales histories provide evidence that the magnitude of the realized path exceeds the threshold in Corollary 4. ■

The refinement discussed in Remark 5 addresses the second concern above about overconservatism. It also illustrates another price of leaving the risk profile unspecified. The difficulty in that case is then not only to guarantee robustness in the worst case, but also to avoid paying for unnecessary robustness when the realized path is easier than anticipated. If preferences move only mildly, then the strategy has to somehow recognize that frequent restarts are unnecessary.

More generally, every easier regime in which the retailer wants good performance can be viewed, in hindsight, as a risk profile that could have been planned for. To perform well across such regimes without selecting one upfront, the strategy has to incorporate additional experimentation capable of distinguishing among them. In that sense, leaving the risk profile unspecified does not remove the planning burden; it shifts that burden into the design of a strategy with more sophisticated experimentation.

7 Numerical experiments

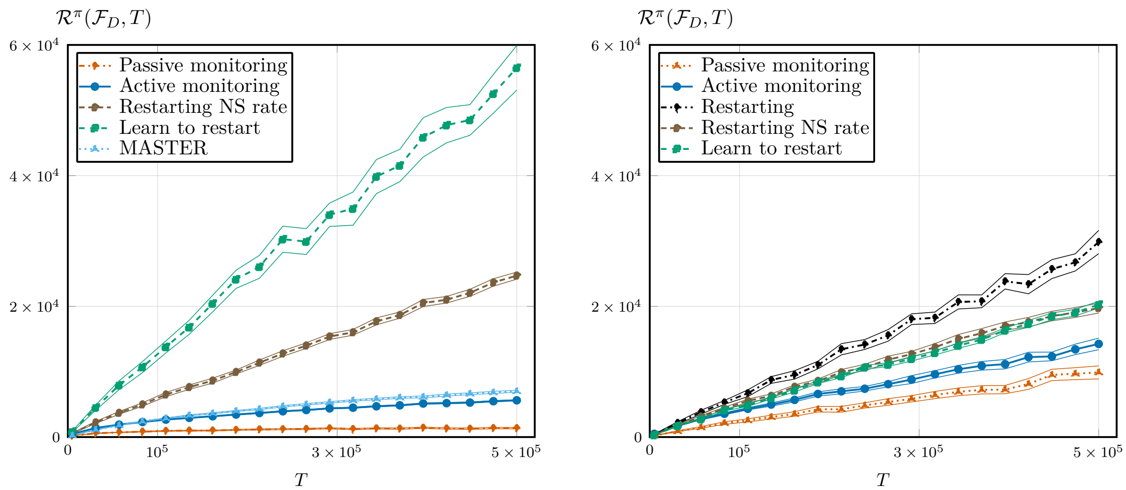
We examine the numerical performance of the robust-planning strategies associated with the three risk profiles studied in Section 5. The experiments have two objectives: to assess how these strategies perform both within and beyond the profiles they were designed to be robust against, and to understand how their performance depends on the downstream policy. To enable a comparison with the existing literature, we focus on MNL preferences, for which MASTER (Wei and Luo 2021) can be implemented with the policy of Foussoul et al. (2023) as its downstream policy. The comparison is conservative for our strategies because MASTER exploits structural inputs from its downstream policy that are generally unavailable for existing dynamic assortment planning policies; recall Section 2. In our experiments, MASTER is thus best understood as a strategy specialized for MNL preferences, whereas our strategies treat the downstream policy as arbitrary and can accommodate broader choice models.

We run each robust-planning strategy with two downstream policies: (*i*) the upper-confidence-bound (UCB) policy $\mathcal{A}^{\mathrm{UCB}}$ of Foussoul et al. (2023), which permits a comparison with MASTER using the same downstream policy; and (*ii*) the Thompson sampling (TS) policy $\mathcal{A}^{\mathrm{TS}}$ of Agrawal et al. (2026), as described in their Algorithm 2, whose near-stationary robustness is not known and for which no MASTER implementation is currently available. This design allows us to separate the effect of our robust-planning strategies on performance from that of the downstream policy.

We do not use the paths of preferences underlying the price-of-robustness results in Section 5. Instead, we construct paths that simply satisfy the defining properties of each risk profile, without seeking out the most adversarial ones within that profile. The numerical experiments are intended to complement the robustness guarantees in Sections 5 and 6 by examining how the strategies perform on less adversarial paths of preferences. Implementation details are provided in Appendix A.

7.1 Passively detectable abrupt changes

We consider paths of preferences in $\mathcal{F}_D$ with a passively detectable abrupt change (Section 5.2). Figure 2 reports the regret of the different strategies, denoted by $\mathcal{R}^\pi(\mathcal{F}_D, T)$ with an abuse of notation. Figures 2a and 2b report the results obtained with the policies $\mathcal{A}^{\mathrm{UCB}}$ and $\mathcal{A}^{\mathrm{TS}}$, respectively.



(a) Downstream policy $\mathcal{A}^{\mathrm{UCB}}$ from Foussoul et al. (2023)

(b) Downstream policy $\mathcal{A}^{\mathrm{TS}}$ from Agrawal et al. (2026)

Figure 2: **Passively detectable abrupt changes** ($\mathcal{F}_D$)**.** Average regret and confidence intervals over 50 instances from $\mathcal{F}_D(\gamma_T, \varepsilon)$ with $\gamma_T = 1/128$ and $\varepsilon = 0.01$, up to $T = 5 \times 10^5$ customers. Regret is computed for passive monitoring (Algorithm 1), active monitoring (Algorithm 2), restarting (Algorithm 3), learn to restart (Algorithm 4), and MASTER (Wei and Luo 2021). The subroutine $\mathcal{A}$ in the first subfigure is the UCB policy of Foussoul et al. (2023), while the subroutine in the second subfigure is Algorithm 2 of Agrawal et al. (2026).

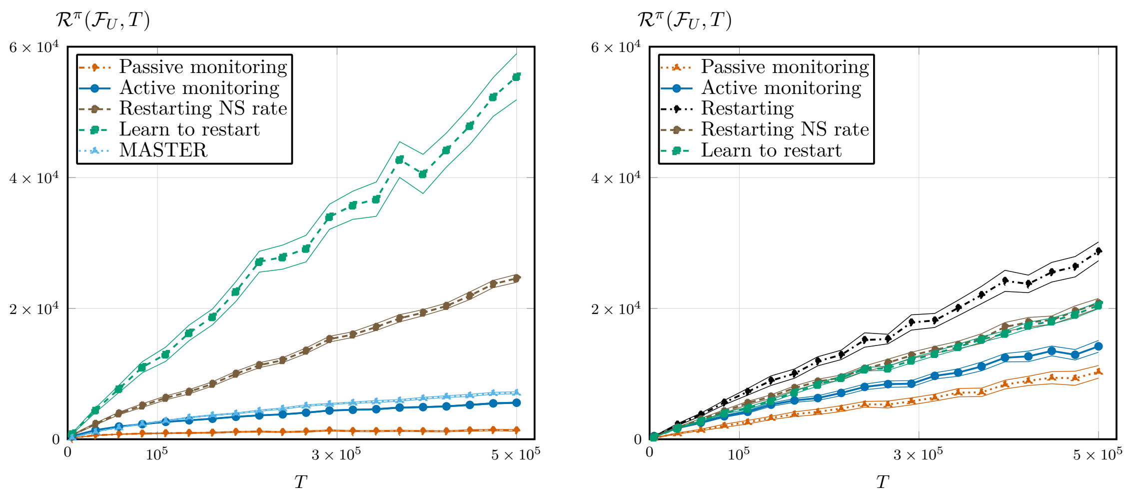
Under $\mathcal{F}_D$, the change affects the purchasing distribution under every assortment, so routine sales can reveal it. Accordingly, the passive monitoring strategy from Algorithm 1 outperforms the other strategies under both downstream policies; see Figures 2a and 2b. Active monitoring from Algorithm 2 also performs well, but incurs more regret because it repeatedly offers assortments from the sentinel family. Passive monitoring instead reuses a single assortment selected by the downstream policy, which may already be revenue-maximizing before the change occurs.

The performance of monitoring strategies depends on the downstream policy around which they are built. Monitoring strategies perform better when built around $\mathcal{A}^{\mathrm{UCB}}$ than when built around $\mathcal{A}^{\mathrm{TS}}$, likely because of the delay between the change and its detection. During this period, the stronger performance of $\mathcal{A}^{\mathrm{UCB}}$ is consistent with its near-stationary robustness under mild departures from static preferences; recall Remark 4. By contrast, if the posterior used by $\mathcal{A}^{\mathrm{TS}}$ has already concentrated before the change, then the policy may continue offering the assortment learned under the old preferences and consequently accumulate regret until the restart.

Beyond monitoring strategies, periodic restarting performs poorly because long periods of static preferences lead it to discard sales information that still represents the preferences of the arriving customers. Among the periodic restarting strategies in Figure 2, the restart frequency for near-stationary robust policies from Corollary 2, marked as the "NS rate," performs better than the more conservative frequency from Corollary 1, which restarts more frequently and is particularly costly during these periods with static preferences. Restarting also performs better with $\mathcal{A}^{\mathrm{TS}}$ than with $\mathcal{A}^{\mathrm{UCB}}$, and the difference is even larger for the learning to restart strategy from Algorithm 4. The role of the downstream policy in determining the cost of repeated restarts becomes particularly clear in Section 7.3.

7.2 Actively detectable abrupt changes

We consider paths of preferences from $\mathcal{F}_U$ with an actively detectable abrupt change (Section 5.3). Figure 3 reports the regret of the different strategies, denoted by $\mathcal{R}^\pi(\mathcal{F}_U, T)$ with an abuse of notation. Figures 3a and 3b report the results obtained with the policies $\mathcal{A}^{\mathrm{UCB}}$ and $\mathcal{A}^{\mathrm{TS}}$, respectively.



(a) Downstream policy $\mathcal{A}^{\mathrm{UCB}}$ from Foussoul et al. (2023)

(b) Downstream policy $\mathcal{A}^{\mathrm{TS}}$ from Agrawal et al. (2026)

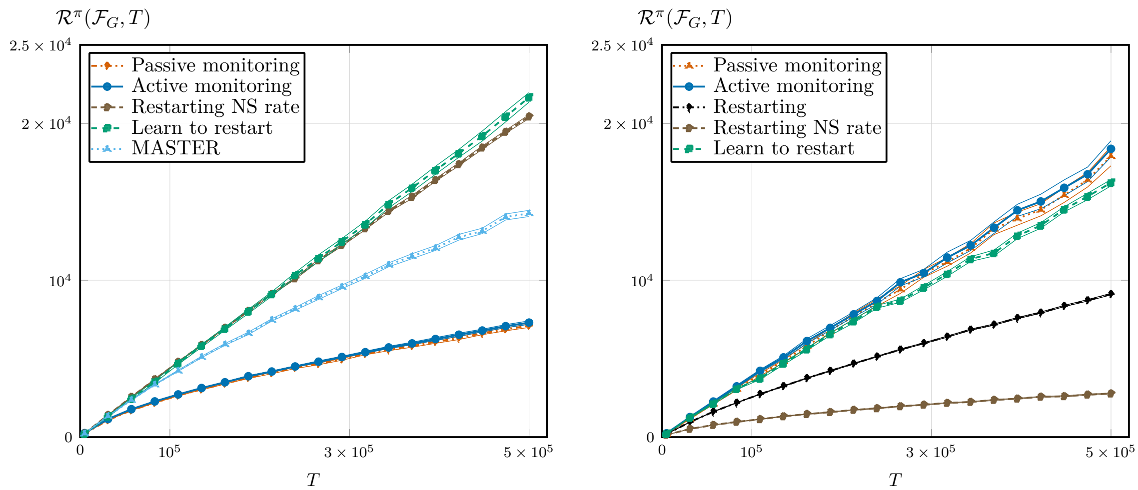
Figure 3: **Actively detectable abrupt changes ($\mathcal{F}_U$).** Average regret and confidence intervals over 50 instances from $\mathcal{F}_U(\gamma_T, \kappa, \phi)$ with $\gamma_T = 1/128$, $\kappa = 0.01$, and $\phi = 105$, up to $T = 5 \times 10^5$ customers. Regret is computed for passive monitoring (Algorithm 1), active monitoring (Algorithm 2), restarting (Algorithm 3), learn to restart (Algorithm 4), and MASTER (Wei and Luo 2021). The subroutine $\mathcal{A}$ in the first subfigure is the UCB policy of Foussoul et al. (2023), while the subroutine in the second subfigure is Algorithm 2 of Agrawal et al. (2026).

The results for actively detectable abrupt changes in Figure 3 are close to those obtained for passively detectable changes in Section 7.1. Most notably, passive monitoring outperforms active monitoring, even though active monitoring is designed to guarantee robustness over $\mathcal{F}_U$. The reason is that $\mathcal{F}_U$ contains paths of different difficulty. Some changes can be revealed only through targeted assortments, whereas others remain visible under many assortment choices. Active monitoring protects against the former, but this protection is costly for the latter.

In the instances considered here, many assortments are informative about the change. The random assortment selected by the passive monitoring strategy, together with the randomization of the downstream policy, provides enough opportunities to detect it. Active monitoring instead tests assortments from a sentinel family covering all products; recall Remark 3. Such experimentation is required for paths in $\mathcal{F}_U$ where routine sales may be uninformative, but imposes an unnecessary cost on the paths considered here. Figure 3 essentially illustrates the cost of *overconservatism* for the retailer: protecting against the more difficult paths in a risk profile can be costly when the path encountered is easier.

7.3 Evolving preferences with controlled magnitude

We consider paths of preferences from $\mathcal{F}_G$ whose magnitude is controlled by a budget (Section 5.4). Figure 4 reports the regret of the different strategies, denoted by $\mathcal{R}^\pi(\mathcal{F}_G, T)$ with an abuse of notation. Figures 4a and 4b report the results obtained with the policies $\mathcal{A}^{\mathrm{UCB}}$ and $\mathcal{A}^{\mathrm{TS}}$, respectively.



(a) Downstream policy $\mathcal{A}^{\mathrm{UCB}}$ from Foussoul et al. (2023)

(b) Downstream policy $\mathcal{A}^{\mathrm{TS}}$ from Agrawal et al. (2026)

Figure 4: **Evolving preferences with controlled magnitude ($\mathcal{F}_G$).** Average regret and confidence intervals over 50 instances from $\mathcal{F}_G(\gamma_T, M_T)$ for some γ_T and some M_T, up to $T = 5 \times 10^5$ customers. Regret is computed for passive monitoring (Algorithm 1), active monitoring (Algorithm 2), restarting (Algorithm 3), learn to restart (Algorithm 4), and MASTER (Wei and Luo 2021). The subroutine $\mathcal{A}$ in the first subfigure is the UCB policy of Foussoul et al. (2023), while the subroutine in the second subfigure is Algorithm 2 of Agrawal et al. (2026).

Figure 4 reveals how the performance of robust-planning strategies depends on the downstream policy around which they are built. With $\mathcal{A}^{\mathrm{UCB}}$, passive and active monitoring outperform nearly all alternatives (Figure 4a), whereas with $\mathcal{A}^{\mathrm{TS}}$, the periodic restarting strategy performs particularly well (Figure 4b). This discrepancy reflects two different costs of responding to evolving preferences: relearning the revenue-maximizing assortment after a restart and continuing to rely on older sales information that may be less informative about the preferences of the arriving customers.

Restarting repeatedly incurs the initial experimentation cost of the downstream policy. Under static preferences, the retailer must typically collect enough observations across different combinations of products to estimate customers' preferences before reliably identifying the revenue-maximizing assortment. Because only products included in the offered assortment generate choice information and capacity limits how many can be offered simultaneously, this initial learning requires experimenting across different assortments, some of which may generate low revenue and, in turn, large regret.

The advantage of periodic restarting with $\mathcal{A}^{\mathrm{TS}}$ relative to $\mathcal{A}^{\mathrm{UCB}}$ is consistent with the empirical evidence that TS policies outperform UCB policies under static preferences (Agrawal et al. 2026), and restarting amplifies the difference. The lowest regret is achieved with the frequency from Corollary 2, while the more conservative frequency from Corollary 1 also performs well. The comparison again illustrates the cost of *overconservatism*: a restart frequency calibrated to guarantee robustness may induce more restarts than a particular path requires, repeatedly incurring the initial experimentation cost.

The same dependence on the downstream policy naturally extends to the learning to restart strategy from Algorithm 4, since its candidate arms are periodically restarted copies of that policy. Accordingly, it performs poorly with $\mathcal{A}^{\mathrm{UCB}}$, only slightly worse than periodic restarting, indicating that the additional cost of learning the restart frequency is small in this case. By contrast, it performs relatively well with $\mathcal{A}^{\mathrm{TS}}$, remaining competitive even with the more specialized MASTER strategy. The difference is notable because MASTER requires an optimism certificate in each period (Wei and Luo 2021), whereas learning to restart treats the downstream policy as a black box.

Monitoring strategies face the opposite trade-off. They retain the accumulated sales history until there is sufficient evidence that preferences have changed, avoiding the repeated initial experimentation cost of restarting but potentially relying on observations collected under older preferences. This trade-off particularly favors $\mathcal{A}^{\mathrm{UCB}}$, whose near-stationary robustness captures its reliability under mild departures from static preferences; recall Remark 4. By contrast, if the posterior used by $\mathcal{A}^{\mathrm{TS}}$ has concentrated, then gradual changes in preferences may leave the policy offering assortments learned from earlier sales for an extended period, even when other assortments would generate larger revenue under the preferences of the arriving customers. This distinction explains why monitoring performs particularly well with $\mathcal{A}^{\mathrm{UCB}}$ but poorly with $\mathcal{A}^{\mathrm{TS}}$ on the paths considered here.

The point is not that monitoring strategies are robust to $\mathcal{F}_G$. They are not, and our adversarial regret analysis identifies paths in $\mathcal{F}_G$ on which they may fail. The numerical results instead show that, over the horizons considered here, monitoring strategies combined with a near-stationary robust policy can still perform well on some paths outside their target profiles. Hence, monitoring strategies should not be dismissed solely because their robustness guarantees apply to narrower risk profiles.

8 Concluding remarks

8.1 Planning for evolving customers' preferences

Evolving customers' preferences can quietly erode a retailer's revenue over time. Much like in the boiling frog apologue, this risk may remain hidden unless it is actively recognized and anticipated. Retailers can address this risk through upfront planning. By articulating a class of evolving preferences against which robustness is sought, the retailer defines a risk profile that governs the revenue it can attain and the structure of the appropriate response. Expanding the risk profile gives the retailer protection against a broader class of evolving preferences, but also raises the price of robustness relative to a clairvoyant retailer. Yet, simple restart-based strategies, with restart timing tied to the retailer's risk profile, provide predictable and near-optimal responses to control this cost.

Our results indicate that robust planning is less about designing increasingly sophisticated dynamic assortment planning strategies and more about deciding which changes in preferences deserve protection. Adaptive approaches reduce the need for retailers to specify a risk profile upfront, but they do so by embedding that choice into the strategy itself. Achieving robustness to evolving preferences through adaptive approaches does not remove the retailer's planning burden. It transfers that burden from a transparent choice of risk profile to the design of a strategy that must use additional experimentation to infer, from a family of candidate profiles, which one best matches the customers it faces.

8.2 Selecting the appropriate response

Selecting a risk profile is the point at which the retailer decides how the remembering-forgetting trade-off is resolved. Under static preferences, older observations remain valuable because each new customer interaction improves estimation. Under evolving preferences, older observations remain valuable only while they still describe the purchasing decisions of the arriving customers. The risk profile specifies the preferences against which the retailer seeks robustness, and therefore specifies the type of sales-information depreciation the retailer plans for before choosing a response.

The risk profile then points to the appropriate response. Such responses can build on policies originally designed for static preferences. They are not limited to periodic restarts. When the risk profile contains only abrupt changes, the retailer can use sales data to decide whether the observations used to estimate customers' preferences should be discarded and the policy restarted. If the change alters the purchasing distribution under every feasible assortment, then routine sales may provide enough evidence that past information has lost relevance. Otherwise, the retailer must actively devote some selling periods to more informative assortments whose role is to reveal whether preferences have changed.

While strategies tailored to a given risk profile perform well when facing paths of preferences from that profile, our numerical experiments indicate that their effectiveness can extend well beyond it. Simple monitoring strategies should therefore not be dismissed solely because they were originally designed to achieve robustness against narrower risk profiles. The appropriate response is indeed not necessarily the most robust one, but the one that best aligns with the retailer's ability to articulate a credible risk profile, its capacity to experiment, and the value it places on operational transparency.

References

Agarwal, A., Luo, H., Neyshabur, B., and Schapire, R. E. (2017). "Corralling a Band of Bandit Algorithms". In: *Proceedings on Learning Theory*. Vol. 65. PMLR, pp. 12–38.

Agrawal, S., Avadhanula, V., Goyal, V., and Zeevi, A. (2019). "MNL-bandit: A dynamic learning approach to assortment selection". In: *Operations Research* 67.5, pp. 1453–1485.

— (2026). "Thompson sampling for the multinomial logit bandit". In: *Mathematics of Operations Research* 51.1, pp. 568–590.

Besbes, O., Gur, Y., and Zeevi, A. (2019). "Optimal exploration-exploitation in a multi-armed bandit problem with non-stationary rewards". In: *Stochastic Systems* 9.4, pp. 319–337.

Besbes, O. and Sauré, D. (2014). “Dynamic Pricing Strategies in the Presence of Demand Shifts”. In: *Manufacturing & Service Operations Management* 16.4, pp. 513–528.

Besbes, O. and Zeevi, A. (2009). “Dynamic pricing without knowing the demand function: Risk bounds and near-optimal algorithms”. In: *Operations Research* 57.6, pp. 1407–1420.

— (2011). “On the minimax complexity of pricing in a changing environment”. In: *Operations Research* 59.1, pp. 66–79.

Blanchet, J., Gallego, G., and Goyal, V. (2016). “A Markov chain approximation to choice modeling”. In: *Operations Research* 64.4, pp. 886–905.

Caro, F. and Gallien, J. (2007). “Dynamic assortment with demand learning for seasonal consumer goods”. In: *Management Science* 53.2, pp. 276–292.

Caro, F., Kök, A. G., and Martínez-de-Albéniz, V. (2020). “The future of retail operations”. In: *Manufacturing & Service Operations Management* 22.1, pp. 47–58.

Cesa-Bianchi, N. and Lugosi, G. (2006). *Prediction, learning, and games*. Cambridge University Press.

Chen, X., Shi, C., Wang, Y., and Zhou, Y. (2021). “Dynamic assortment planning under nested logit models”. In: *Production and Operations Management* 30.1, pp. 85–102.

Cheung, W. C., Simchi-Levi, D., and Zhu, R. (2022). “Hedging the drift: Learning to optimize under nonstationarity”. In: *Management Science* 68.3, pp. 1696–1713.

Cover, T. M. and Thomas, J. A. (2005). *Elements of information theory*. John Wiley & Sons.

Farias, V. F., Jagabathula, S., and Shah, D. (2013). “A nonparametric approach to modeling choice with limited data”. In: *Management Science* 59.2, pp. 305–322.

Feldman, J. and Topaloglu, H. (2015). “Bounding optimal expected revenues for assortment optimization under mixtures of multinomial logits”. In: *Production and Operations Management* 24.10, pp. 1598–1620.

Foster, D. P. and Vohra, R. (1999). “Regret in the On-Line Decision Problem”. In: *Games and Economic Behavior* 29.1-2, pp. 7–35.

Foussoul, A., Goyal, V., and Gupta, V. (2023). “MNL-bandit in non-stationary environments”. In: *arXiv preprint arXiv:2303.02504*.

Gallego, G. and Topaloglu, H. (2014). “Constrained assortment optimization for the nested logit model”. In: *Management Science* 60.10, pp. 2583–2601.

Hannan, J. (1957). “Approximation to Bayes risk in repeated play”. In: *Contributions to the Theory of Games* 3.2, pp. 97–139.

Honhon, D., Jonnalagedda, S., and Pan, X. A. (2012). “Optimal algorithms for assortment selection under ranking-based consumer choice models”. In: *Manufacturing & Service Operations Management* 14.2, pp. 279–289.

Keskin, N. B. and Zeevi, A. (2017). “Chasing demand: Learning and earning in a changing environment”. In: *Mathematics of Operations Research* 42.2, pp. 277–307.

Kök, A. G., Fisher, M. L., and Vaidyanathan, R. (2015). “Assortment planning: Review of literature and industry practice”. In: *Retail Supply Chain Management: Quantitative Models and Empirical Studies*, pp. 175–236.

Lai, T. L. and Robbins, H. (1985). “Asymptotically efficient adaptive allocation rules”. In: *Advances in Applied Mathematics* 6.1, pp. 4–22.

Li, S., Luo, Q., Huang, Z., and Shi, C. (2025). “Online Learning for Constrained Assortment Optimization Under Markov Chain Choice Model”. In: *Operations Research* 73.1, pp. 109–138.

Mahajan, S. and van Ryzin, G. (2001). “Stocking retail assortments under dynamic consumer substitution”. In: *Operations Research* 49.3, pp. 334–351.

Neu, G. (2015). “Explore no more: Improved high-probability regret bounds for non-stochastic bandits”. In: *Advances in Neural Information Processing Systems* 28.

Rooderkerk, R. P., DeHoratius, N., and Musalem, A. (2022). “The past, present, and future of retail analytics: Insights from a survey of academic research and interviews with practitioners”. In: *Production and Operations Management* 31.10, pp. 3727–3748.

Rusmevichientong, P., Shen, Z.-J. M., and Shmoys, D. B. (2010). “Dynamic assortment optimization with a multinomial logit choice model and capacity constraint”. In: *Operations Research* 58.6, pp. 1666–1680.

Sauré, D. and Zeevi, A. (2013). “Optimal dynamic assortment planning with demand learning”. In: *Manufacturing & Service Operations Management* 15.3, pp. 387–404.

Sheth, J. N. (1967). “A review of buyer behavior”. In: *Management Science* 13.12, pp. 718–756.

Talluri, K. and van Ryzin, G. (2004). “Revenue management under a general discrete choice model of consumer behavior”. In: *Management Science* 50.1, pp. 15–33.

Train, K. E. (2009). *Discrete choice methods with simulation*. Cambridge University Press.

van Ryzin, G. and Vulcano, G. (2015). “A market discovery algorithm to estimate a general class of nonparametric choice models”. In: *Management Science* 61.2, pp. 281–300.

Vorotyntseva, Y. and Honhon, D. (2026). “A Behavioral Study of Assortment Planning”. In: *Manufacturing & Service Operations Management* 28.3, pp. 726–741.

Vulcano, G., van Ryzin, G., and Ratliff, R. (2012). “Estimating primary demand for substitutable products from sales transaction data”. In: *Operations Research* 60.2, pp. 313–334.

Wei, C.-Y. and Luo, H. (2021). “Non-stationary reinforcement learning without prior knowledge: An optimal black-box approach”. In: *Conference on learning theory*. PMLR, pp. 4300–4354.

Online Appendix

(Saving Kermit: Dynamic Assortment Planning in a Boiling Market)

A Computational setup for Section 7

We use the same setup across all numerical experiments. There are $N = 10$ products, a cardinality constraint $K = 4$, and unit profits $w_i = 1$ for all $i \in \mathcal{N}$. We consider paths of MNL preferences $F^{(\mathbb{N})} \in \mathcal{F}_{\text{MNL}}$. At time t, a path $F^{(\mathbb{N})}$ is represented by its vector of attraction parameters $\nu^t \equiv \nu^t(F^{(\mathbb{N})}) := (\nu_i^t : i \in \mathcal{N})$. For an assortment $S \in \mathcal{S}$, the purchasing probabilities are $p_i(S, F^t) = \nu_i^t(1 + \sum_{j \in S} \nu_j^t)^{-1}$ for $i \in S$, and $p_0(S, F^t) = (1 + \sum_{j \in S} \nu_j^t)^{-1}$. Since all profits are one, the assortment $S^*(F^t)$ selected by the clairvoyant retailer at time t contains the K products with the largest attraction parameters. For a strategy π, we compute its *pseudo regret* using expected rather than realized revenues as $\sum_{t=1}^{T} \left(r(S^*(F^t), F^t) - r(S^{\pi,t}, F^t)\right)$.

For each risk profile and horizon T, we generate 50 independent paths of preferences and evaluate every strategy on the same paths. For each strategy, we compute the average pseudo regret $\bar{R}$ across these paths together with its 95% confidence interval, $\bar{R} \pm 1.96\, s/\sqrt{50}$, where s is the sample standard deviation of the 50 pseudo regret values. The horizon grid contains 20 evenly spaced values from 5,000 to 500,000. The active monitoring strategy from Algorithm 2 uses the sentinel family $\mathcal{E} = \{\{1, 2, 3, 4\}, \{5, 6, 7, 8\}, \{9, 10\}\}$, which forms a product cover; recall Remark 3. For the periodic restarting strategy from Algorithm 3, motivated by Corollaries 1 and 2, we set the restart interval to $\Delta = \lceil (NT/M_T)^{1/4} \rceil$ for an arbitrary downstream policy and $\Delta = \lceil (NT/M_T)^{1/3} \rceil$ for a near-stationary robust policy, with M_T specified separately for each experiment below.

A.1 Instance construction for Section 7.1

The experiment uses instances in $\mathcal{F}_{D,\text{MNL}} \subseteq \mathcal{F}_D$, as defined below. We fix $\gamma_T = 1/128$ and $\varepsilon = 0.01$. For each horizon T, we sample the change time τ uniformly from $\{\lfloor T/4 \rfloor, \dots, \lceil 3T/4 \rceil\}$. We then set $\nu^t = \nu^a$ for $t < \tau$ and $\nu^t = \nu^b$ for $t \geq \tau$, where ν^a and ν^b are specified below. For the periodic restarting strategy in Algorithm 3, we use $M_T = 105$ to determine the restart frequencies.

(Instance construction) We consider MNL preferences in $\mathcal{F}_{\text{MNL}}$ with $N = 10$ products and capacity $K = 4$. Let $I_1 = \{1, 2, 3, 4\}$, $I_2 = \{5, 6, 7, 8\}$, and $I_3 = \{9, 10\}$. Fix $\gamma_T \in (0, 1/64)$, set $h = (24\gamma_T)^{-1}$, and let $q = h/(h+2)$. Fix $\varepsilon \in (0, \log(1 + h/2)/4)$. Choose $\eta \in (0, 1/12)$ such that $\eta \leq \gamma_T/4$, $d_{\text{bin}}(q, 4\eta) \geq \varepsilon$, and $d_{\text{bin}}(4\eta, q) \geq \varepsilon$, where $d_{\text{bin}}(x, y) = x \log(x/y) + (1-x) \log((1-x)/(1-y))$. Such a choice exists because $d_{\text{bin}}(q, 4\eta) \to +\infty$ and $d_{\text{bin}}(4\eta, q) \to \log(1/(1-q)) = \log(1 + h/2) > \varepsilon$ as $\eta \downarrow 0$. In the experiments, we determine a feasible value of η numerically. Finally, fix $\lambda = \eta \exp(-4(\varepsilon + 1)/\eta)/8$. We define $\mathcal{F}_{D,\text{MNL}} \subseteq \mathcal{F}_{\text{MNL}}$ as the set of MNL preferences paths with change time $\tau \geq 2$, attraction vector ν^a before τ, and attraction vector ν^b from τ onward, where $\nu_i^a = h\mathbf{1}\{i \in I_1\} + \eta\mathbf{1}\{i \in I_2 \cup I_3\}$ and $\nu_i^b = \eta\mathbf{1}\{i \in I_1\} + h\mathbf{1}\{i \in I_2\} + \lambda\mathbf{1}\{i \in I_3\}$, with $\nu_0^a = \nu_0^b = 1$. Under the conditions above, one can verify that $\mathcal{F}_{D,\mathrm{MNL}} \subseteq \mathcal{F}_D(\gamma_T, \varepsilon)$.

A.2 Instance construction for Section 7.2

The experiment uses instances in $\mathcal{F}_{U,\mathrm{MNL}} \subseteq \mathcal{F}_U$, as defined below. We fix $\gamma_T = 1/128$, $\kappa = 0.01$, and $\phi = 105$. For each horizon T, we sample the change time τ uniformly from $\{\lfloor T/4 \rfloor, \ldots, \lceil 3T/4 \rceil\}$. We then set $\nu^t = \nu^a$ for $t < \tau$ and $\nu^t = \nu^b$ for $t \geq \tau$, where ν^a and ν^b are specified below. For the periodic restarting strategy in Algorithm 3, we use $M_T = \phi = 105$ to determine the restart frequency, as in the passively detectable experiment above in Appendix A.1.

(Instance construction) We consider paths of MNL preferences in $\mathcal{F}_{\mathrm{MNL}}$ with $N = 10$ products and capacity $K = 4$. Let $I_1 = \{1, 2, 3, 4\}$, $I_2 = \{5, 6, 7, 8\}$, and $I_3 = \{9, 10\}$. Fix $\gamma_T \in (0, 1/64)$, set $h = (24\gamma_T)^{-1}$, and let $\eta = \gamma_T/4$. Fix $\kappa > 0$ and $\phi > \kappa$ such that $\kappa < (1+6\gamma_T)^{-1}\log((6\gamma_T^2)^{-1}) + \log((1+\gamma_T)/(1+(6\gamma_T)^{-1}))$ and $\phi > \log((6\gamma_T^2)^{-1}) + \log(1+(6\gamma_T)^{-1})$.

We define $\mathcal{F}_{U,\mathrm{MNL}} \subseteq \mathcal{F}_{\mathrm{MNL}}$ as the set of paths of MNL preferences with one change at $\tau \geq 2$, vector of attraction parameters ν^a before τ, and vector of attraction parameters ν^b from τ onward, where $\nu_i^a = h\mathbf{1}\{i \in I_1\} + \eta\mathbf{1}\{i \in I_2 \cup I_3\}$ and $\nu_i^b = \eta\mathbf{1}\{i \in I_1 \cup I_3\} + h\mathbf{1}\{i \in I_2\}$, with $\nu_0^a = \nu_0^b = 1$. Under the conditions above, one can verify that $\mathcal{F}_{U,\mathrm{MNL}} \subseteq \mathcal{F}_U(\gamma_T, \kappa, \phi)$. In contrast with the passively detectable construction, the products in $I_3 = \{9, 10\}$ keep the same attraction throughout the selling horizon. In particular, we select ϕ so that the conditions defining $\mathcal{F}_U$ are satisfied and set $M_T = \phi = 105$ to determine the restart frequency of the periodic restarting strategy.

A.3 Instance construction for Section 7.3

The experiment considers paths of preferences $F^{(\mathbb{N})}$ in $\mathcal{F}_G(\gamma_T, M_T)$. Since sinusoidal paths are commonly used to represent non-stationarity in the MAB literature (Besbes et al. 2019), we construct paths of MNL preferences by letting their attraction parameters vary sinusoidally over time, as specified below. The resulting paths are parametrized by $\tilde{M}_T$. Importantly, $\tilde{M}_T$ is a parameter of the construction and does not necessarily coincide with the magnitude of the resulting path.

(Instance construction) We use $\tilde{M}_T = 50$ in Section 7.3 and repeat the experiment below with $\tilde{M}_T = 5000$. Let $c_N = (N-1)/2$. For product $i \in \{1, \ldots, N\}$, define the deterministic baseline $\mu_i = 0.20(c_N - (i-1))/c_N$. Then, we draw $U \sim \mathrm{Unif}[0, 2\pi]$, $\psi_i \sim \mathrm{Unif}[0, 2\pi]$, and $a_i \sim \mathrm{Unif}[0.85, 1.15]$. For product i, we define $\varphi_i := (2\pi(i-1)/N + U) \bmod 2\pi$. Next, we introduce $c_T = 1 + 0.25(\tilde{M}_T)^{1/2}$, and fix $\omega = 2\pi c_T/T$. The attraction of product i at time t is then defined by:

$$\nu_i^t = \exp\left(\mu_i + 0.55a_i \sin(\omega(t-1) + \varphi_i) + 0.15 \cdot 0.55 \sin(2\omega(t-1) + \psi_i)\right).$$

The baseline μ_i creates persistent differences across products, so the path does not start from symmetric preferences. The common phase U randomizes the initial position of the path, while the product-specific phases $\varphi_i = (2\pi(i-1)/N + U) \bmod 2\pi$ spread products evenly along the oscillation. The loading a_i introduces small product-level differences in the amplitude of the main component.

Hence, larger values of $\tilde{M}_T$ increase c_T, so the attraction vectors complete more oscillations over the same horizon. The main sinusoidal term $0.55 a_i \sin(\omega(t-1) + \varphi_i)$ drives most of the variation in attractions and can change which products are most attractive. The smaller term $0.15 \cdot 0.55 \sin(2\omega(t-1) + \psi_i)$ adds an additional product-specific perturbation.

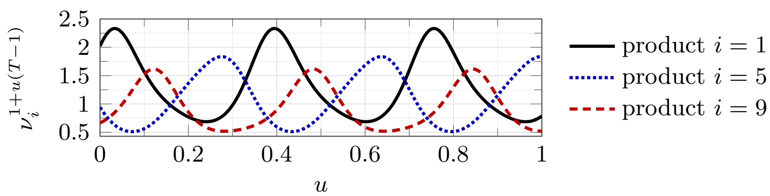
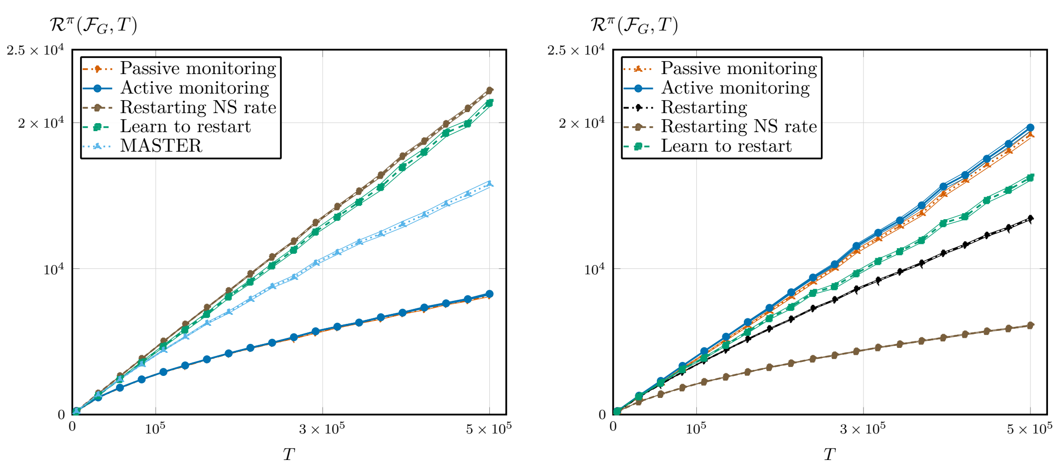


Figure A.1: Attraction parameters used to construct the paths of MNL preferences in Section 7.3 with $\tilde{M}_T = 50$. The figure shows one realization of the construction for three products. The horizontal axis reports normalized time $u = (t-1)/(T-1)$, $t \in [T]$, and the vertical axis reports the corresponding attraction parameter $\nu_i^{1+u(T-1)}$ for $i \in \{1, 5, 9\}$.

Figure A.1 illustrates the resulting instance construction through one realization of the attraction parameters for three products when $\tilde{M}_T = 50$. After generating each path of preferences $F^{(\mathbb{N})}$, we compute its magnitude $\mathcal{M}(F^{(\mathbb{N})}, T)$. The restart frequencies are then tuned using this realized magnitude: for each generated path, we measure $\mathcal{M}(F^{(\mathbb{N})}, T)$ and compute the corresponding restart frequency. We repeat the experiment with $\tilde{M}_T = 5000$. The results, reported in Figure A.2, are consistent with those in Section 7.3, except that the learning to restart strategy performs slightly better than periodic restarting for $\mathcal{A}^{\mathrm{UCB}}$ and remains competitive with MASTER for $\mathcal{A}^{\mathrm{TS}}$.



(a) Downstream policy $\mathcal{A}^{\mathrm{UCB}}$ from Foussoul et al. (2023)

(b) Downstream policy $\mathcal{A}^{\mathrm{TS}}$ from Agrawal et al. (2026)

Figure A.2: **Evolving preferences with controlled magnitude ($\mathcal{F}_G$).** Average regret and confidence intervals over 50 instances from $\mathcal{F}_G(\gamma_T, M_T)$ for some γ_T and some M_T, up to $T = 5 \times 10^5$ customers. Regret is computed for passive monitoring (Algorithm 1), active monitoring (Algorithm 2), restarting (Algorithm 3), learn to restart (Algorithm 4), and MASTER (Wei and Luo 2021). The subroutine $\mathcal{A}$ in the first subfigure is the UCB policy of Foussoul et al. (2023), while the subroutine in the second subfigure is Algorithm 2 of Agrawal et al. (2026).